



영상의학과
김학진



갑상선 영상의학

2. THE THYROID GLAND

3. THE PARATHYROID GLAND



학습목표

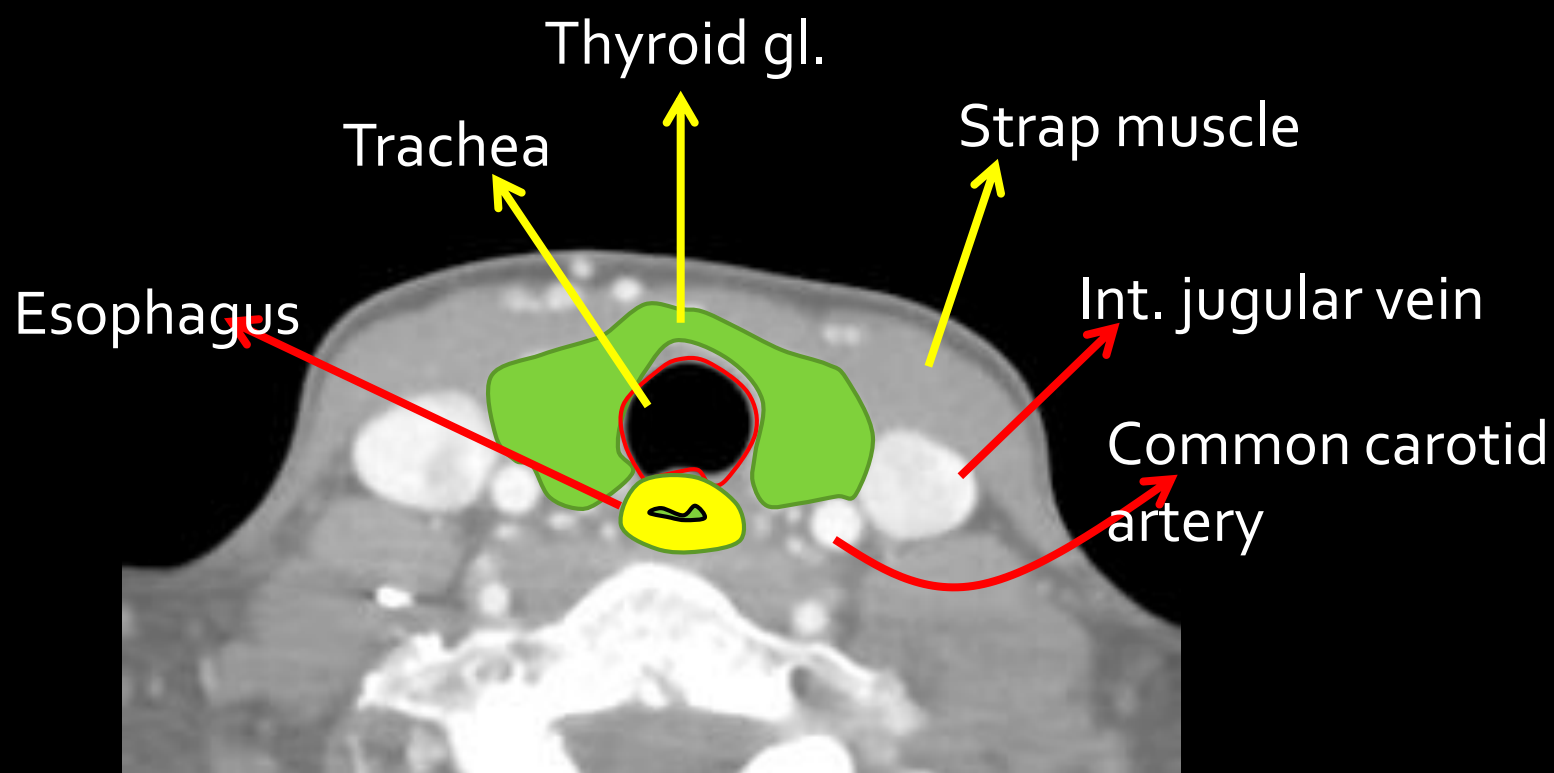
- 갑상선, 부갑상선의 영상해부학 이해
 - Cystic mass와 solid mass의 감별
 - 양성종양과 악성종양의 감별 이해
- 



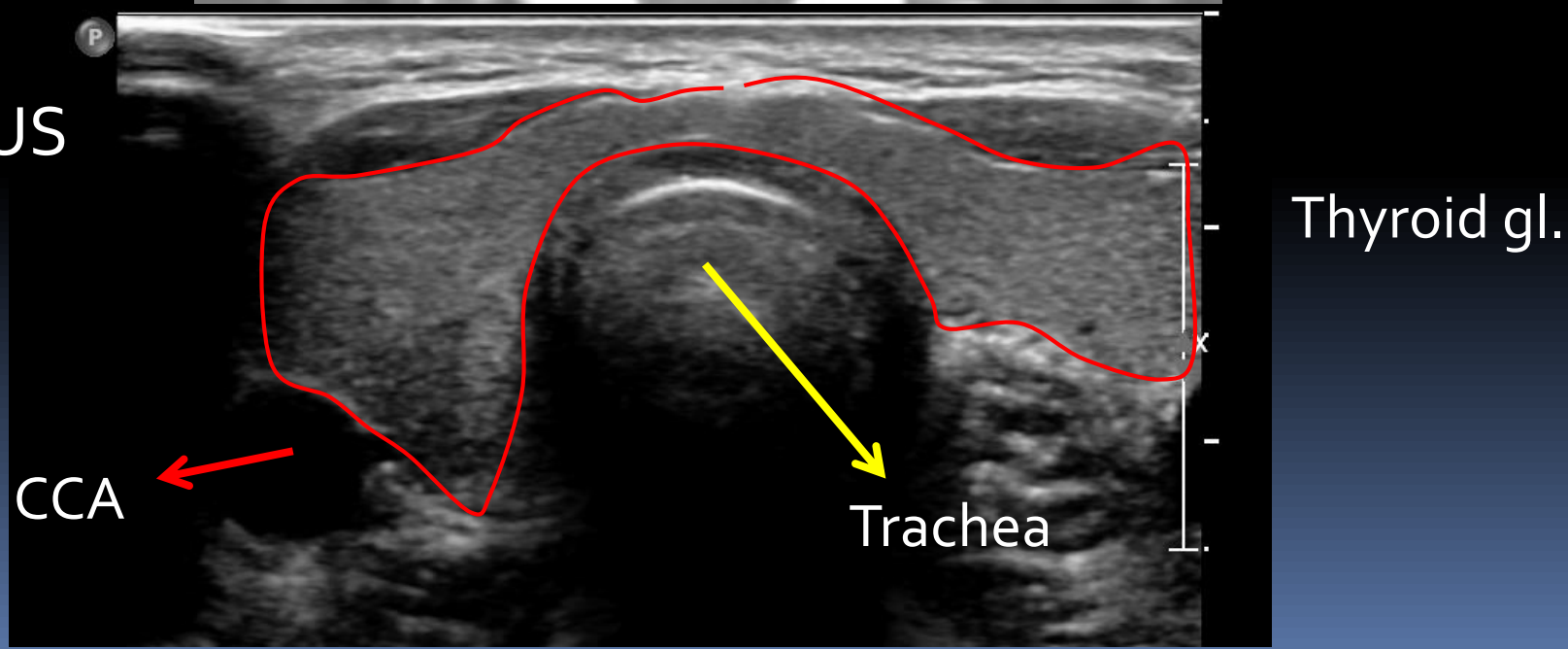
강의 순서

- Thyroid gland
 - Anatomy
 - Focal mass
 - Benign
 - Malignant
 - Diffuse disease
 - Thyroiditis
- Parathyroid gland

CT



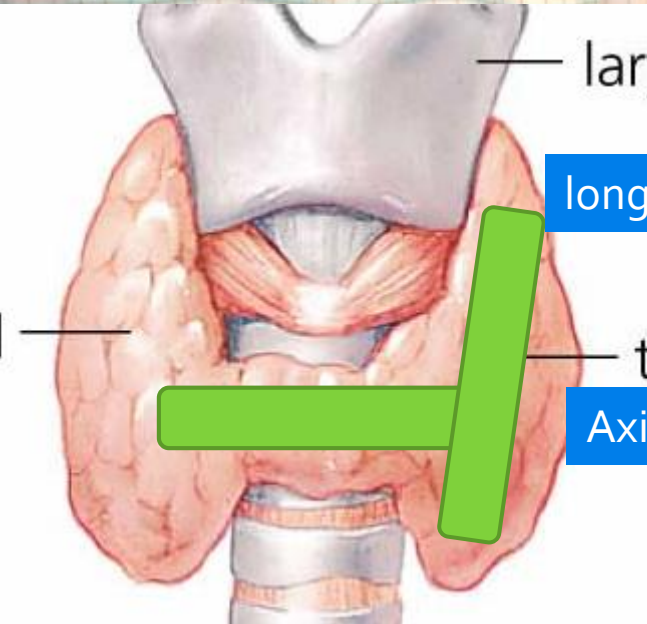
US





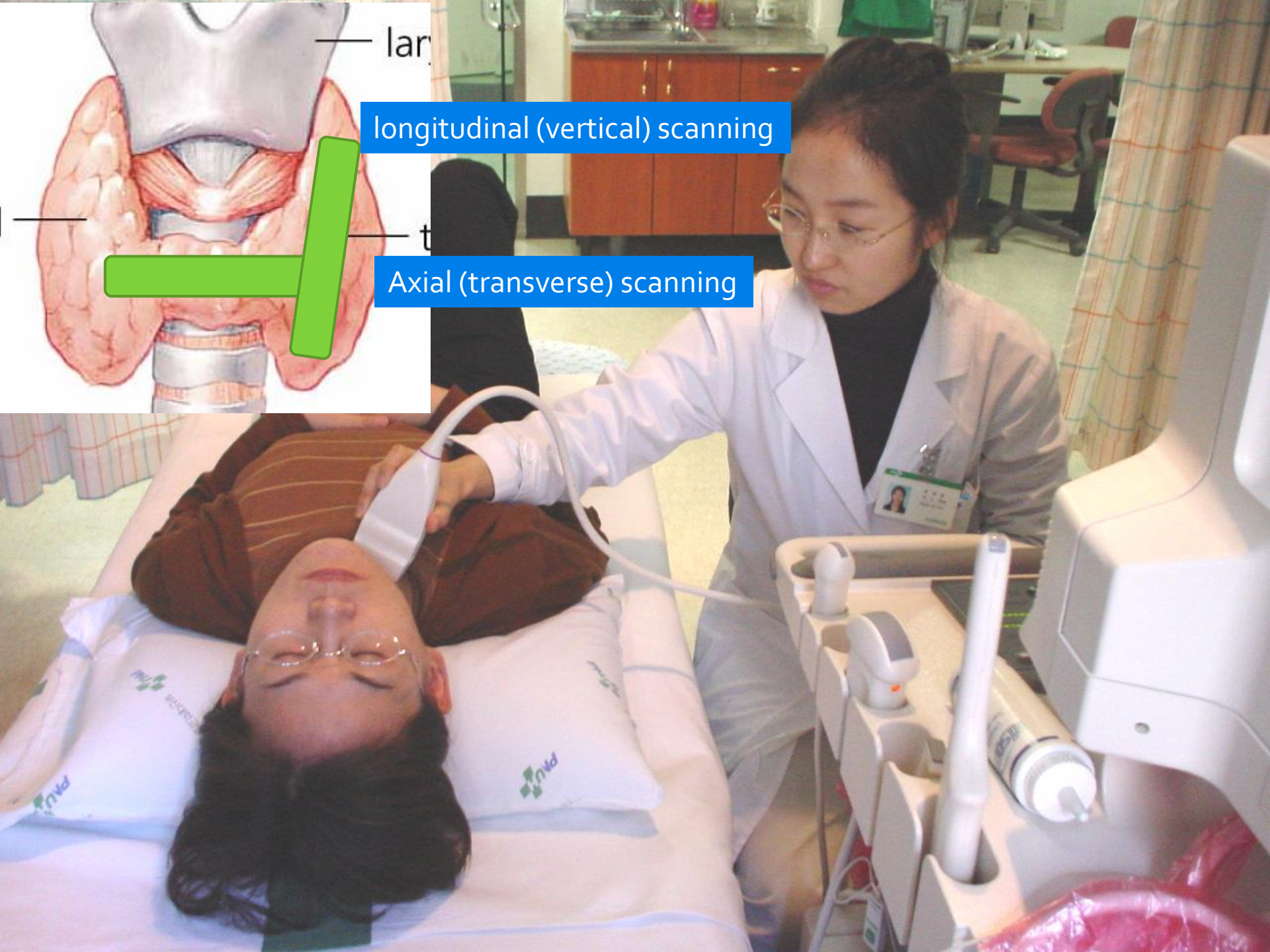
Imaging modalities for the thyroid lesions

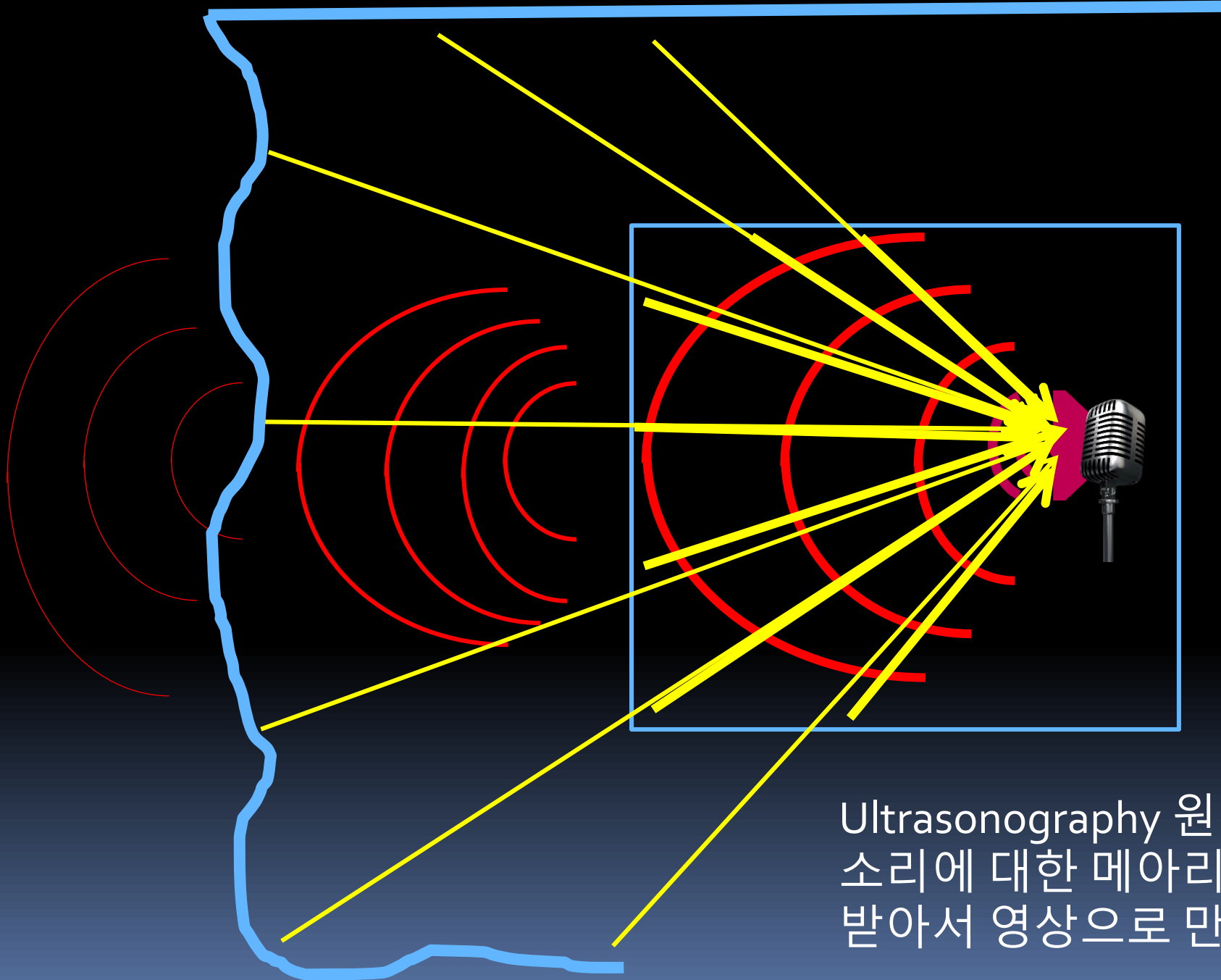
- Ultrasonography
 - CT
 - MRI
 - RI scan
 - PET-CT
- 



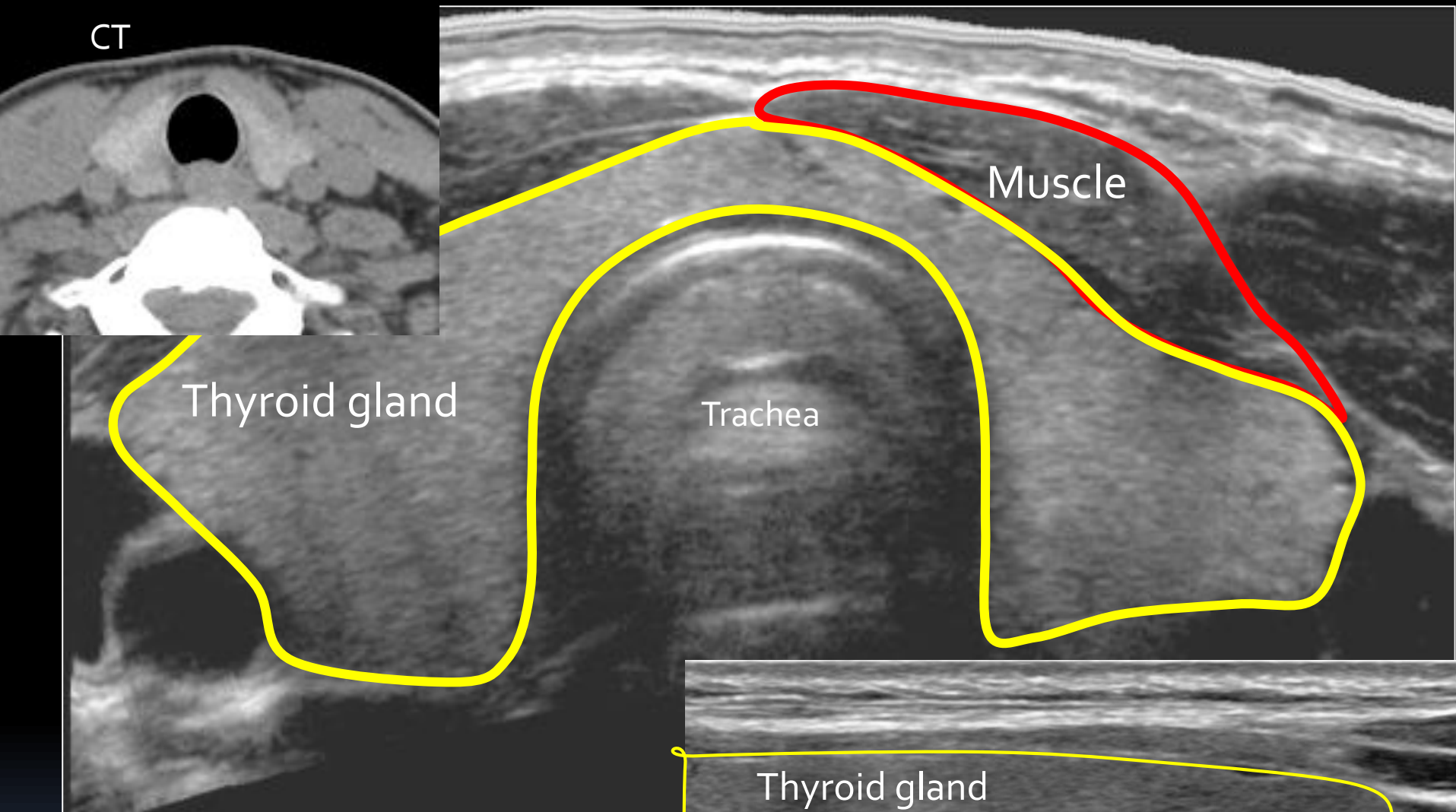
longitudinal (vertical) scanning

Axial (transverse) scanning



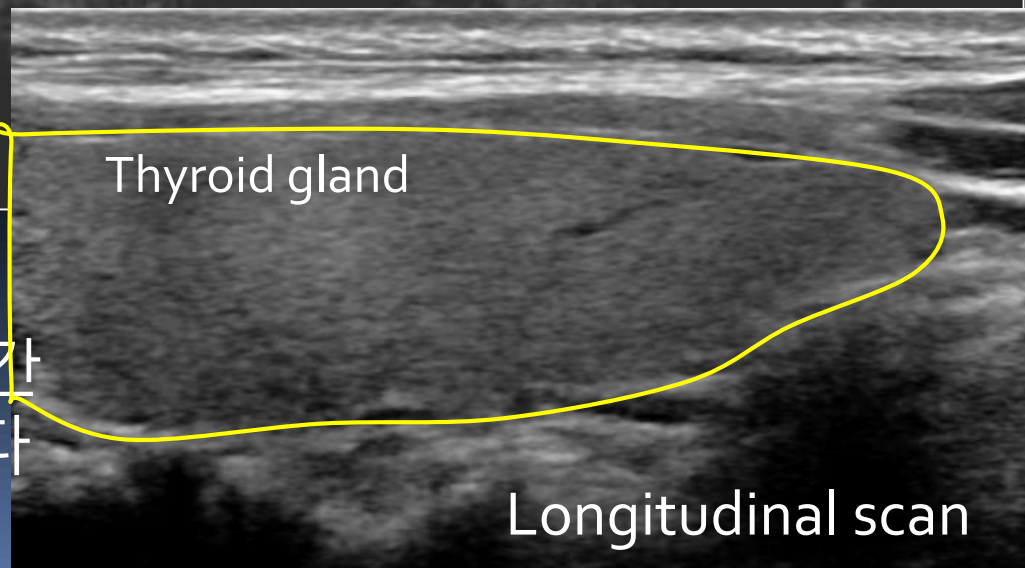


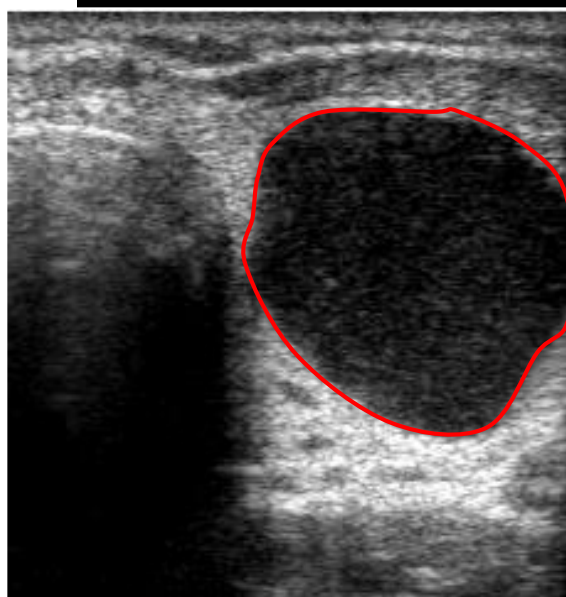
Ultrasonography 원리:
소리에 대한 메아리를
받아서 영상으로 만듦



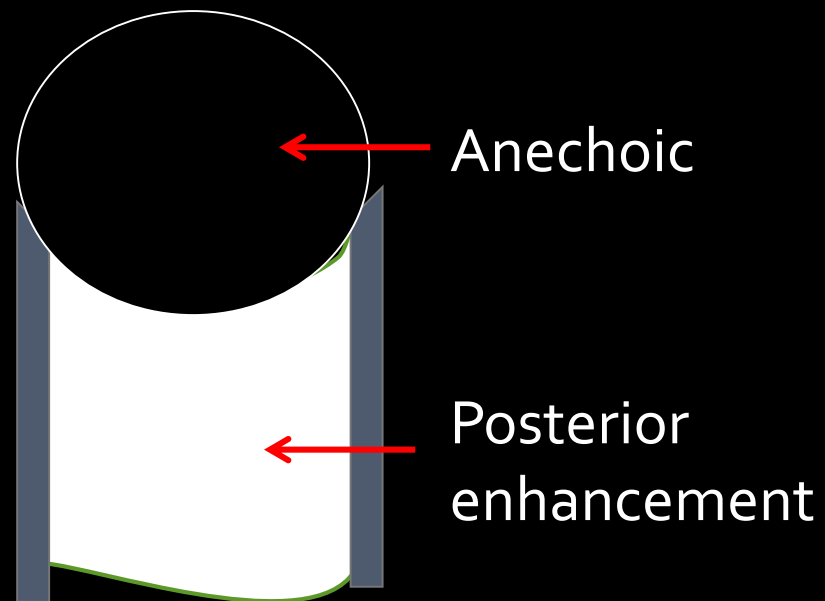
Ultrasonography:

Thyroid는 주변 muscle보다 약간
Hyperechoic하며 조밀(fine)하다



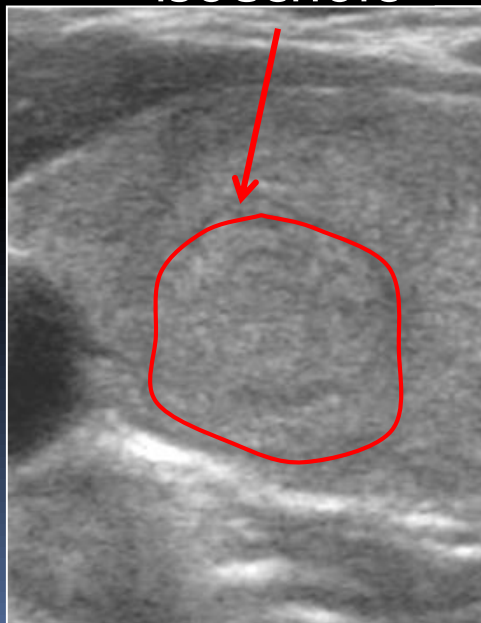


Cystic mass

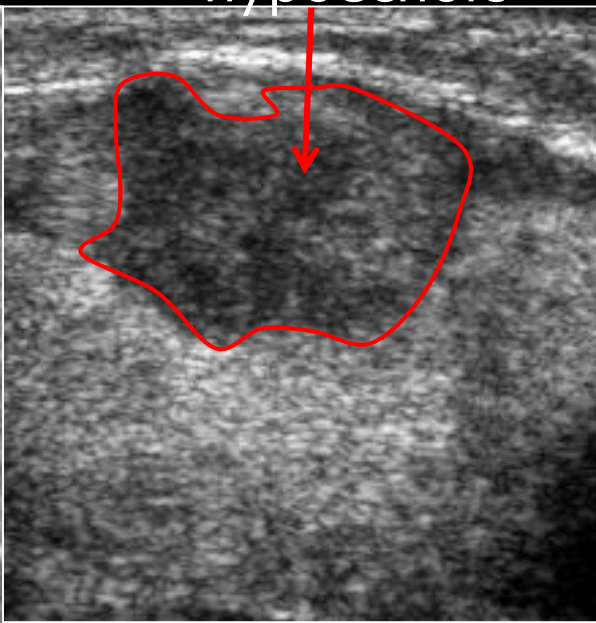


Solid mass

isoechoic



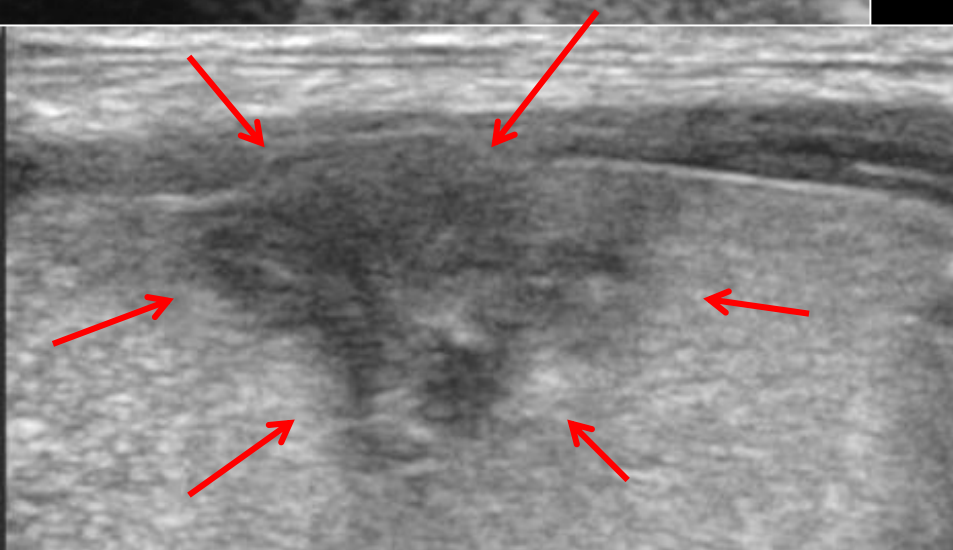
hypoechoic



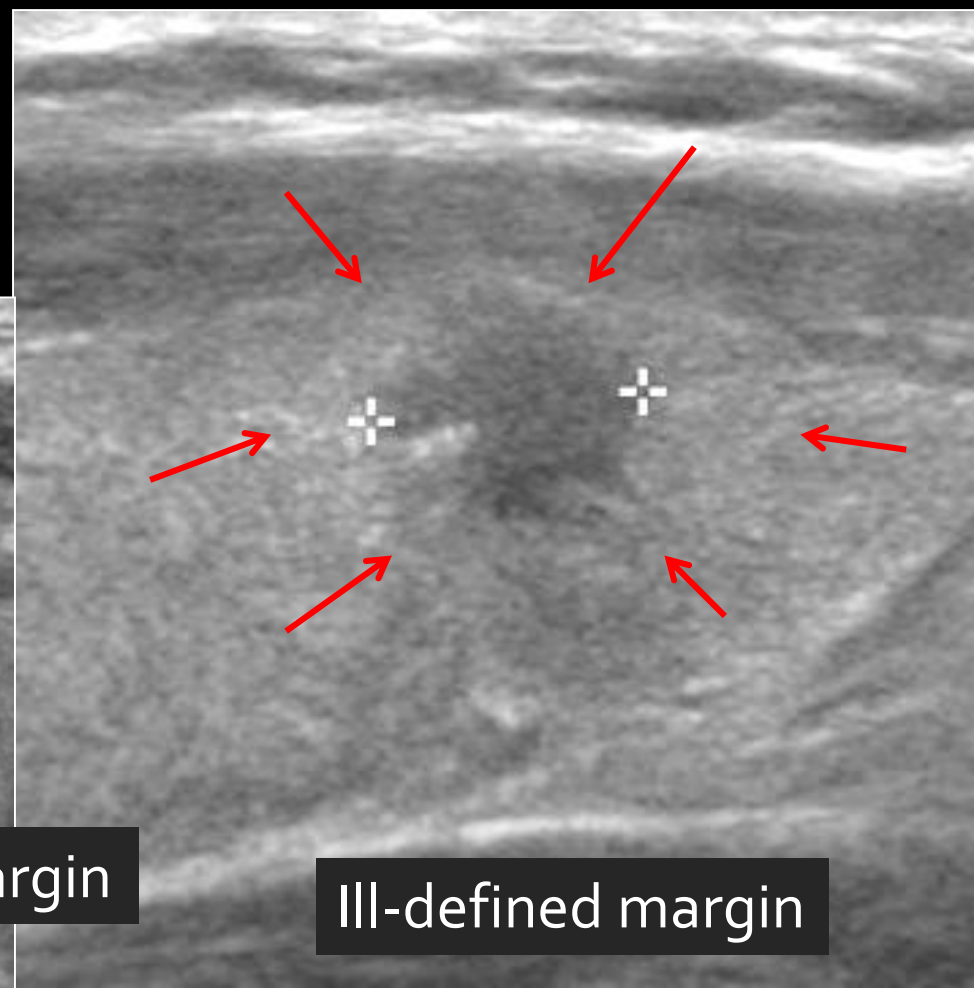
Cystic vs. solid mass

Margin

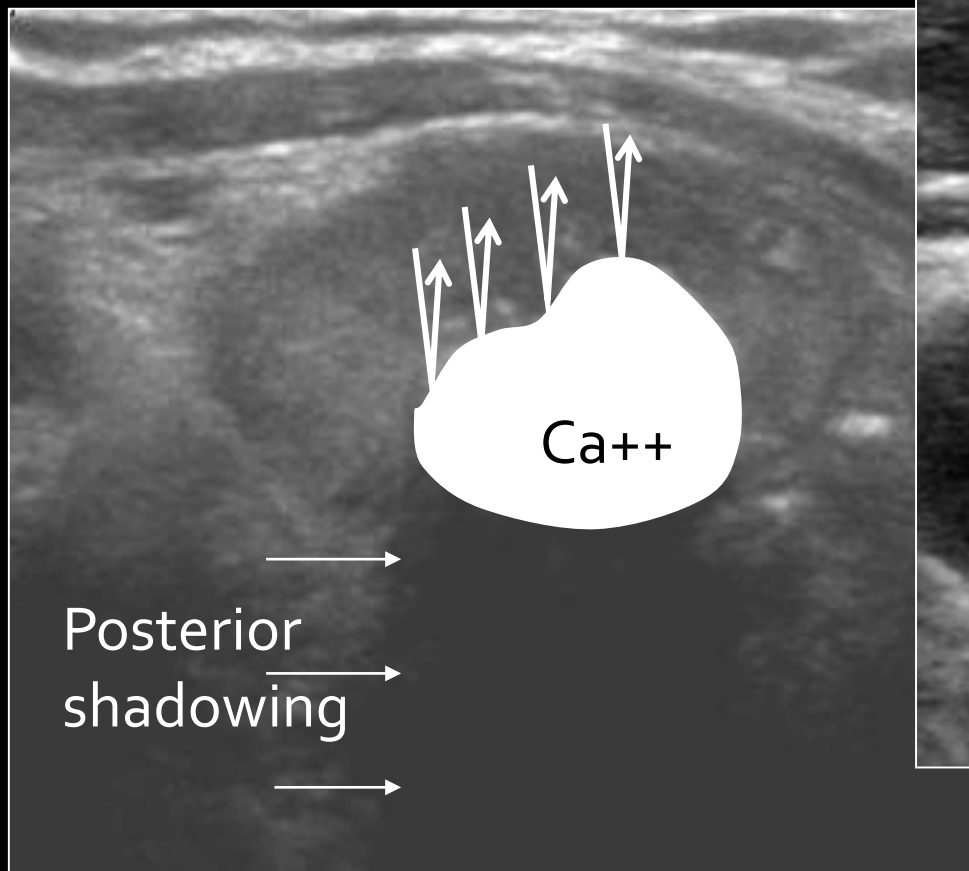
Well-defined margin



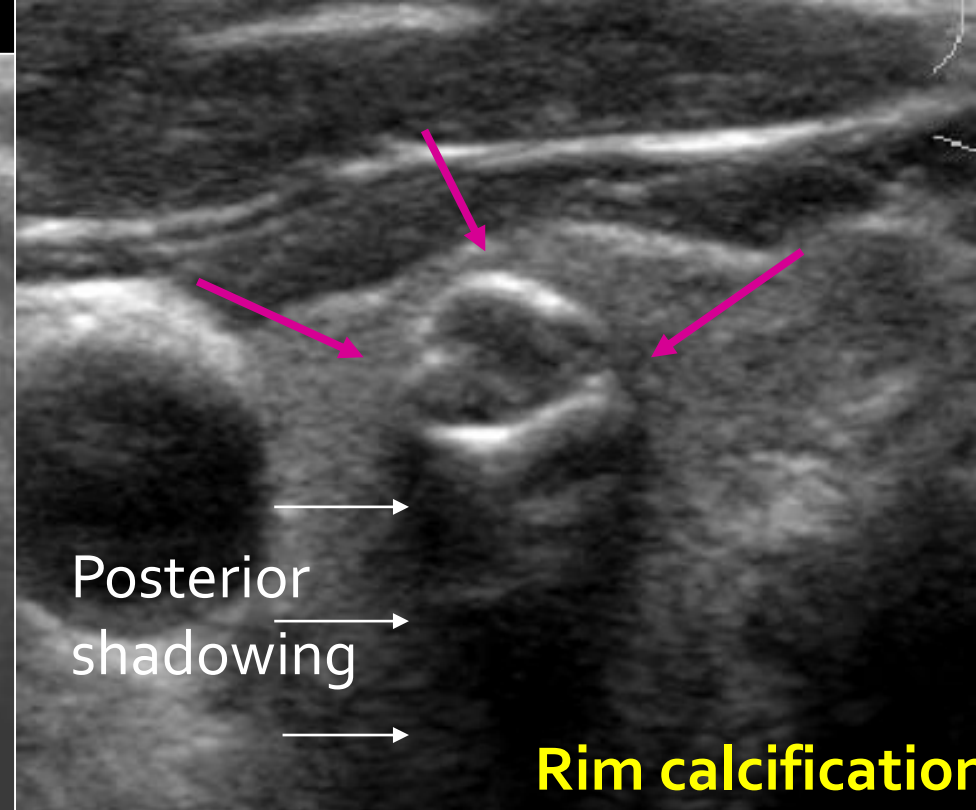
Ill-defined & lobulated margin



Ill-defined margin

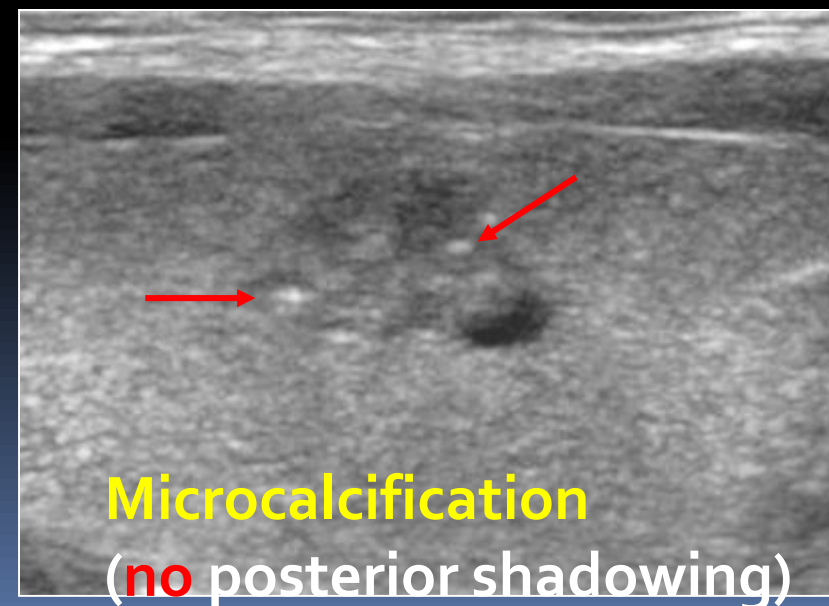


Macrocalcification



Rim calcification

Calcification

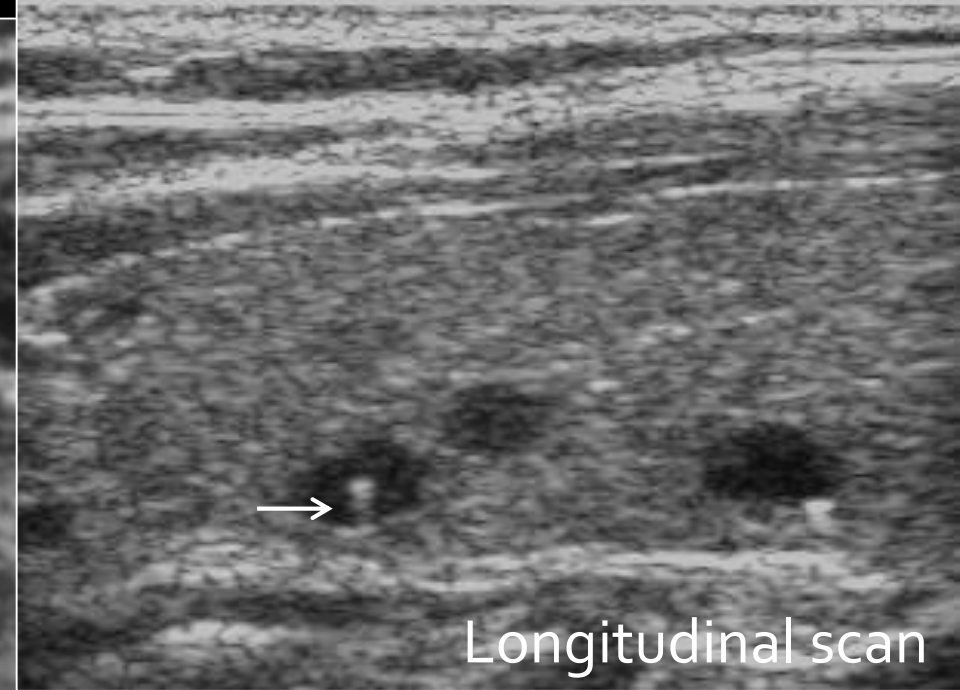
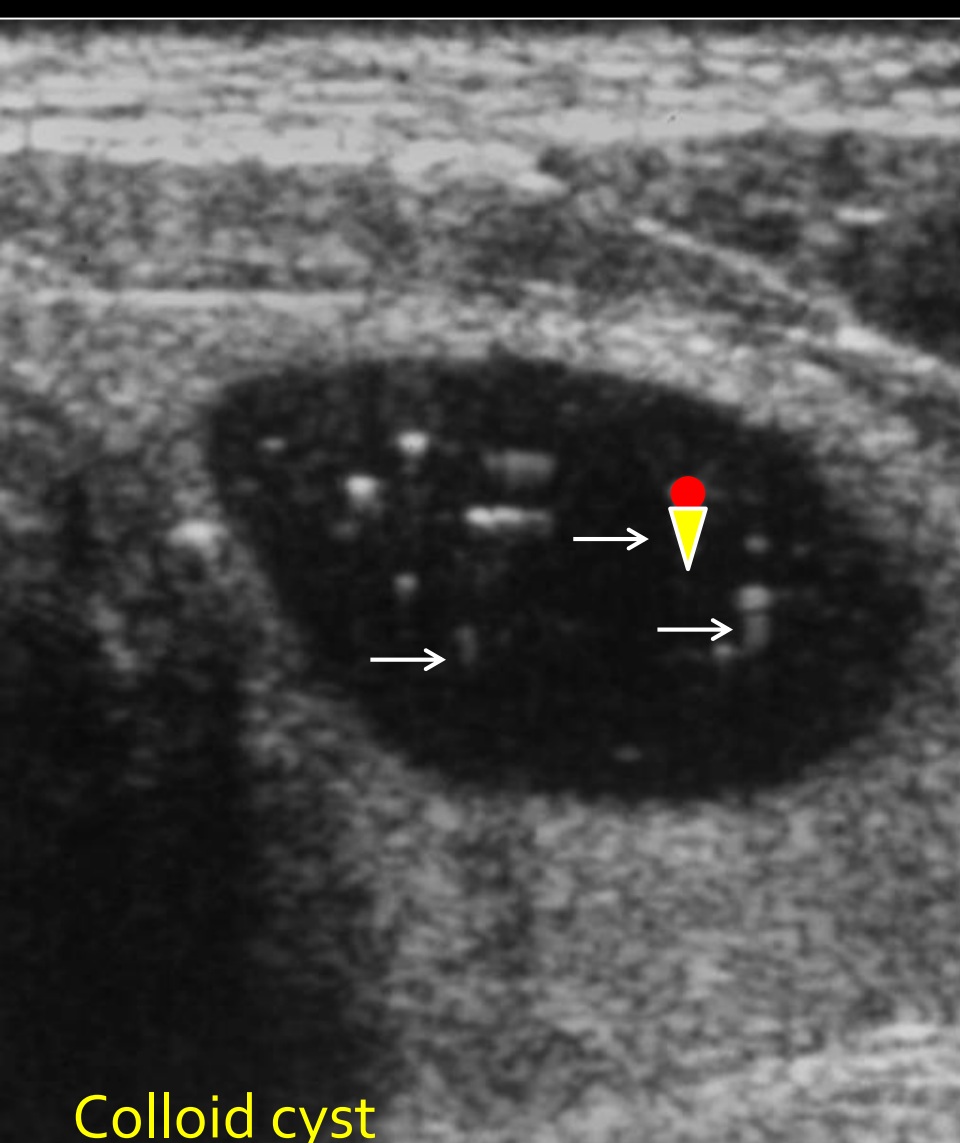


Microcalcification
(no posterior shadowing)



Ultrasonographic features suggesting **benign** lesion

- Thin well-defined halo, regular margin, macrocalcifications
 - Comet-tail artifact within cystic nodule
 - Honeycomb appearance
 - No neck lymphadenopathy
- 



Colloid cyst

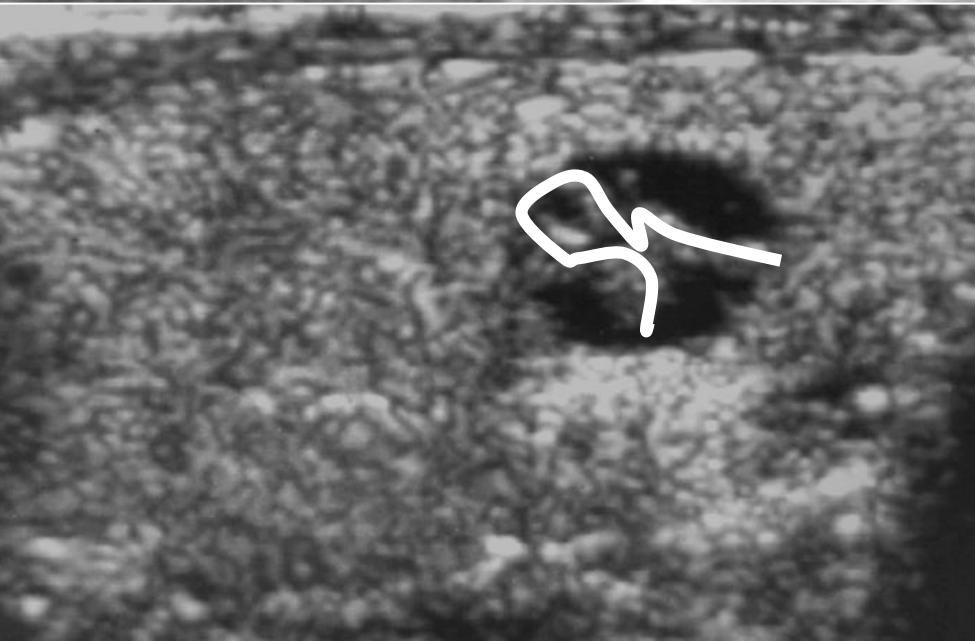
- well-defined anechoic (cystic) mass 내에 Comet tail (유성꼬리)
- Honeycomb (벌집)



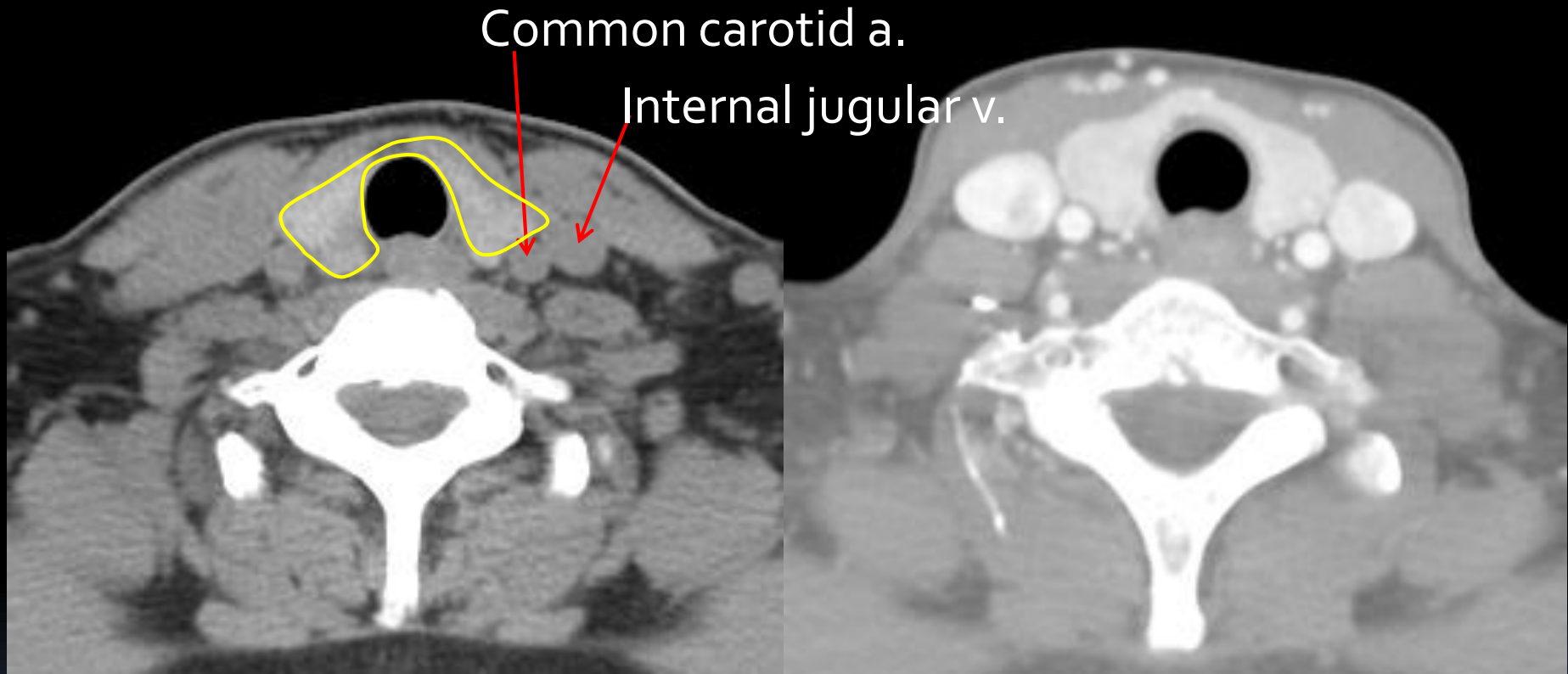


Honeycomb appearance

Colloid cyst



http://hpecpg_thyroid.tripod.com/needlecolloidcyst.htm



Precontrast CT

-Thyroid gland에 있는 iodine 성분; high density를 보임

Postcontrast CT

-조영증강 잘 됨

calcification



Solid

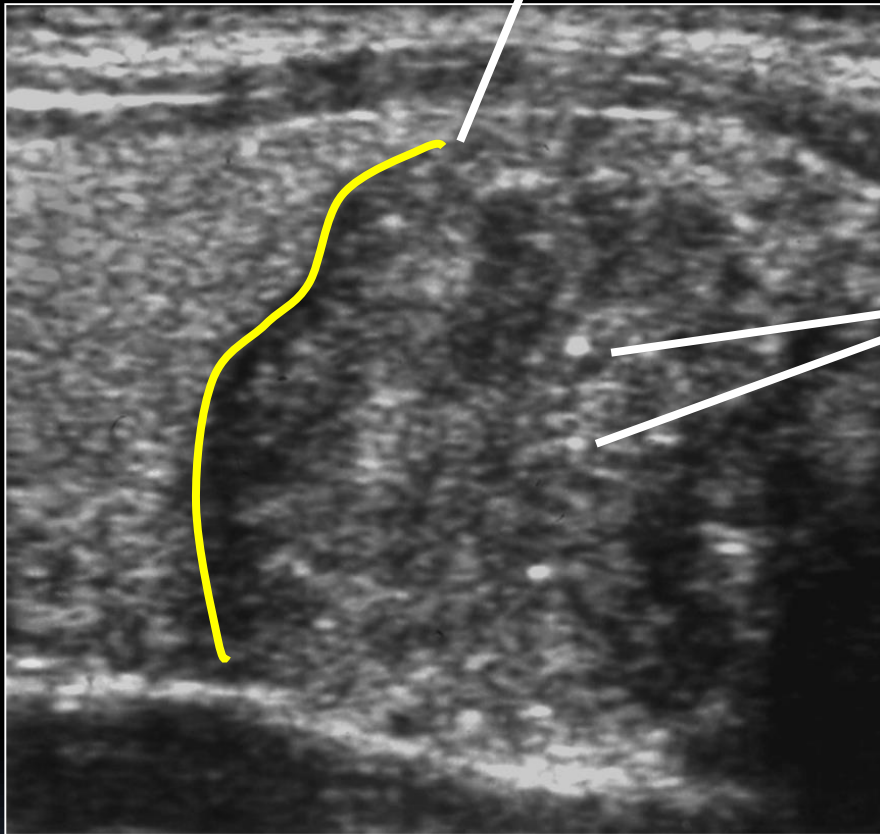


Cyst

- Well-margined diffuse enlargement
- Heterogeneous nodular appearance
 - Cyst; low density (green arrow)
 - Solid; intermediate density (red arrows)
 - calcification; high density (yellow arrows)

Multinodular goiter

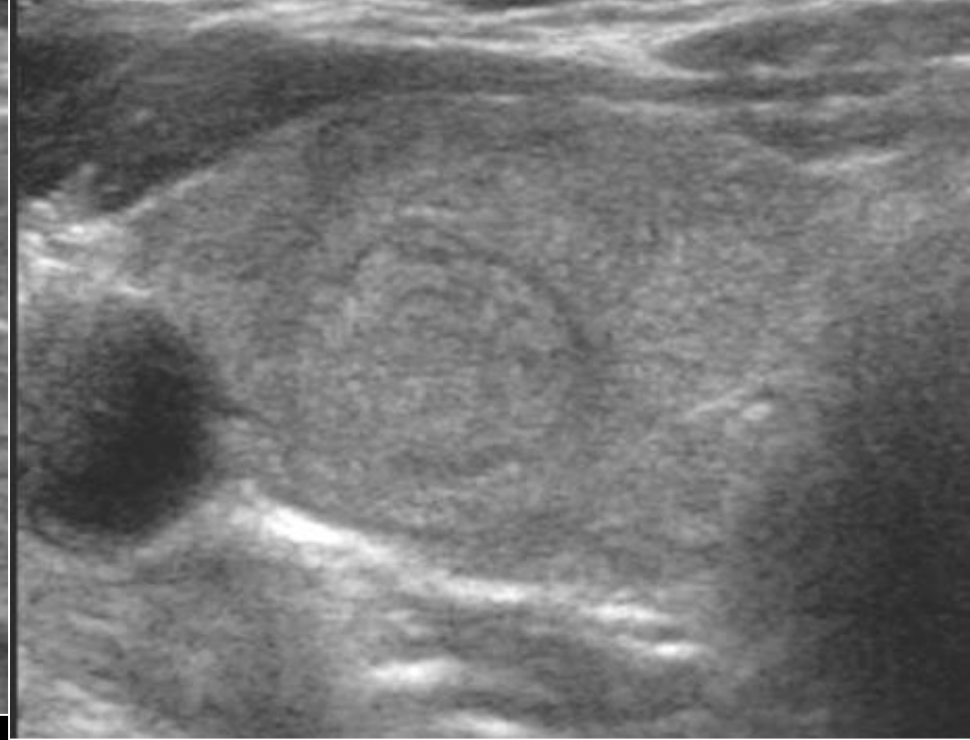
Irregular, lobulated margins



Malignant nodule

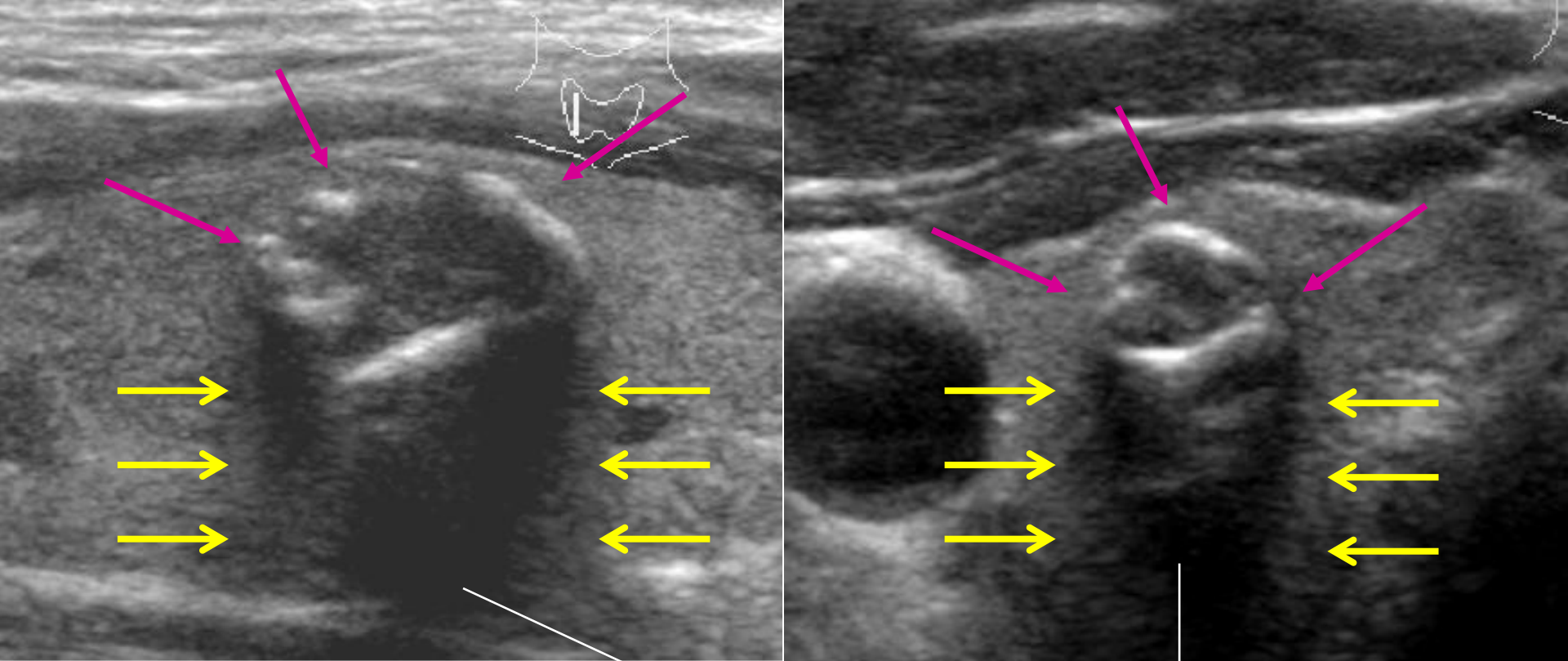
Microcalcification; no shadowing

hypoechoic



- isoechoic center (solid)
- thin continuous peripheral hypoechoic halo

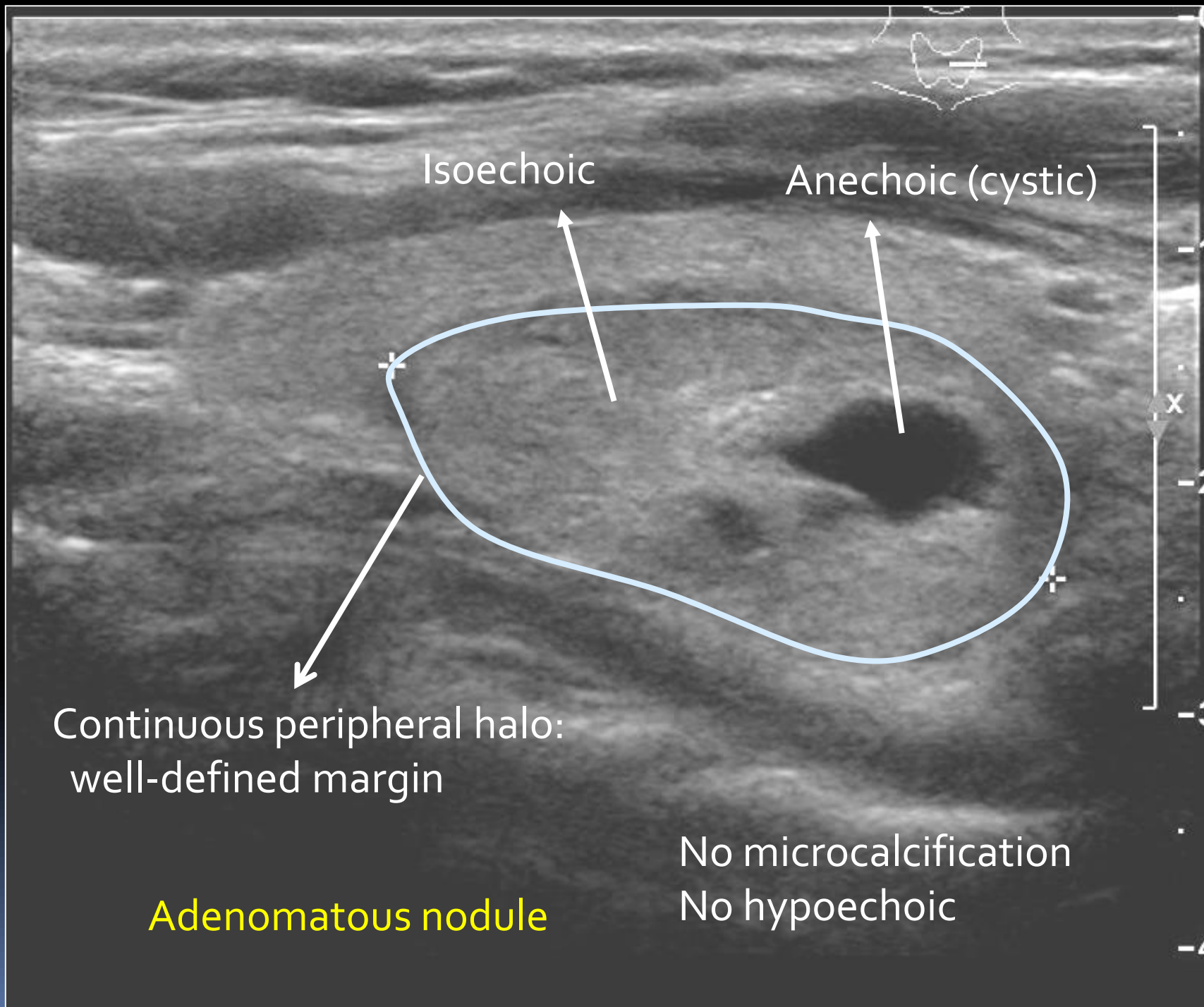
Adenoma



Rim calcification (red arrows)
-macrocalcification
-benign more likely

Posterior shadowing





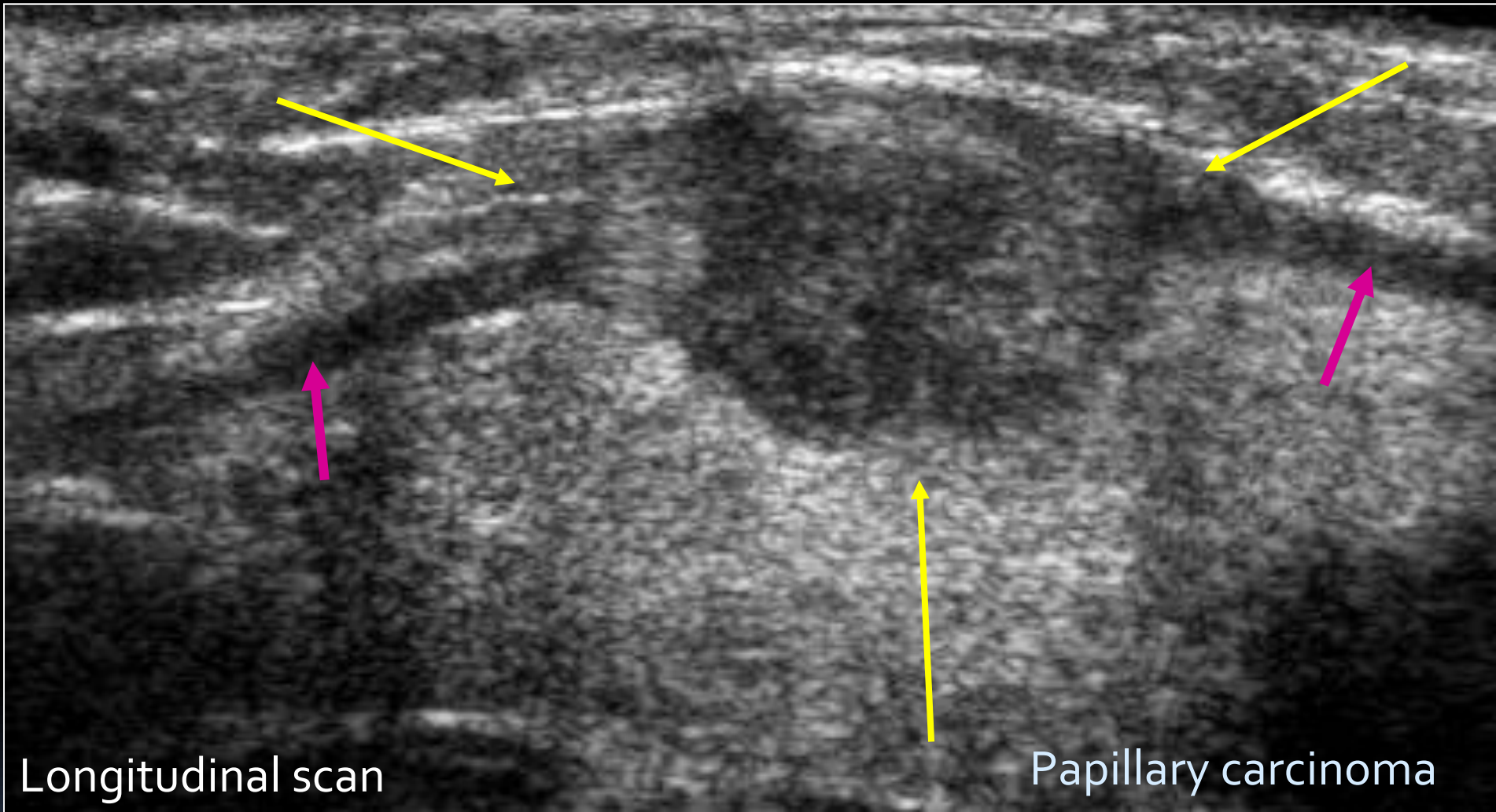


Malignant Thyroid Mass

- Differentiated Ca
 - Medullary Ca
 - Anaplastic Ca
 - Lymphoma
- 

Ultrasonographic features suggesting **malignant** lesion

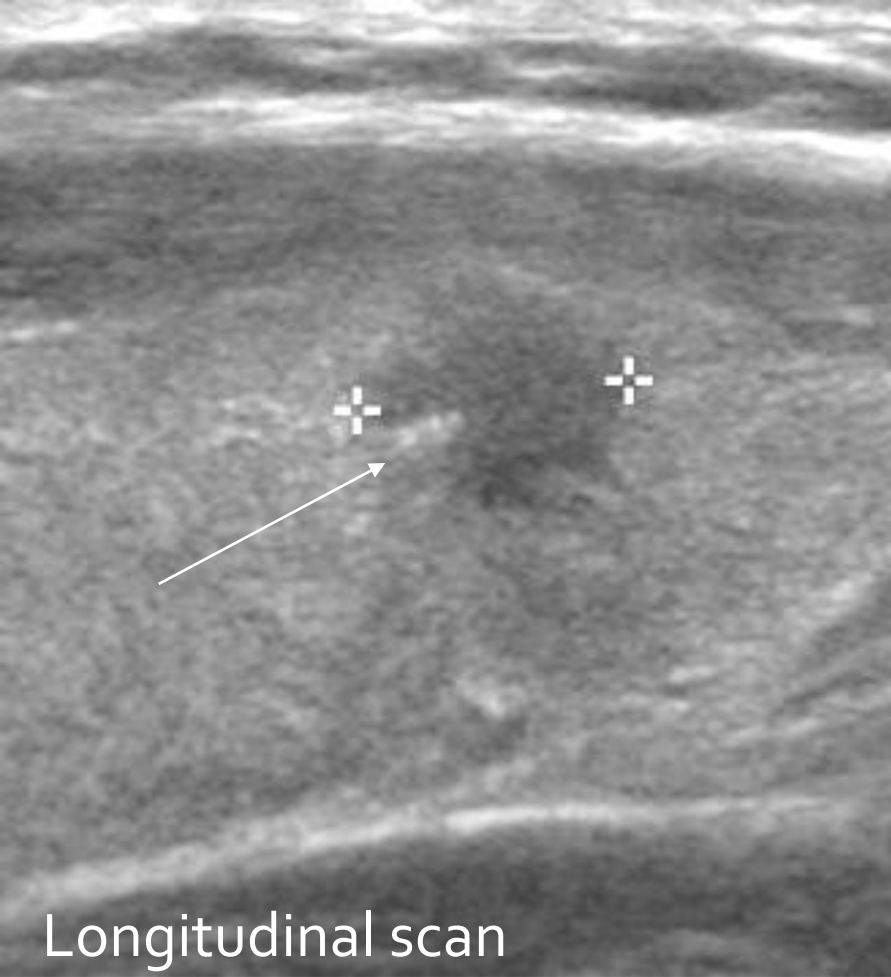
- ⇔ Microcalcification: specificity 94%
- ⇔ lack of cystic change: sensitivity 94.3%
- ⇔ **microcalcification + irregular margin:**
Highest diagnostic accuracy 90.5%
- ⇔ Malignant node



Irregular, lobulated margin (yellow arrows)

Hypoechoic

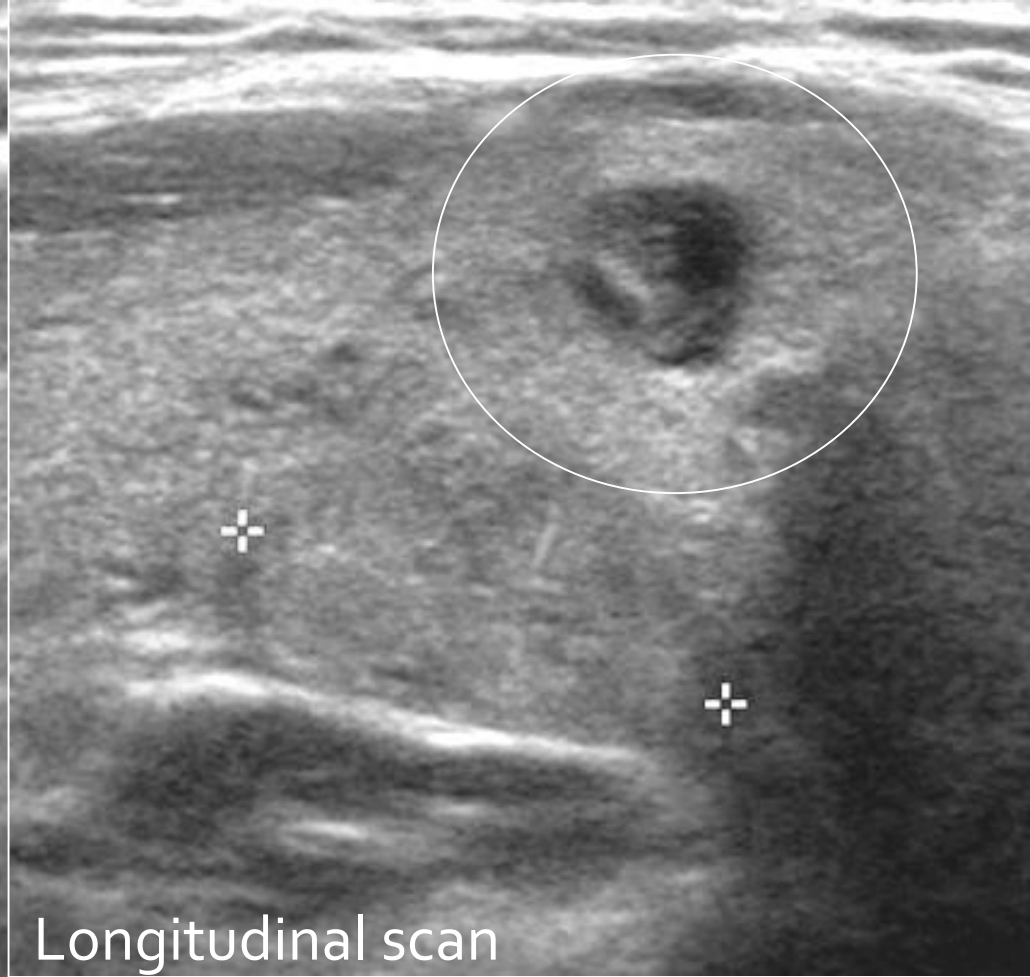
Thyroid capsule involvement (red arrows)



Longitudinal scan

- irregular margin
- hypoechoic
- microcalcification (arrow)

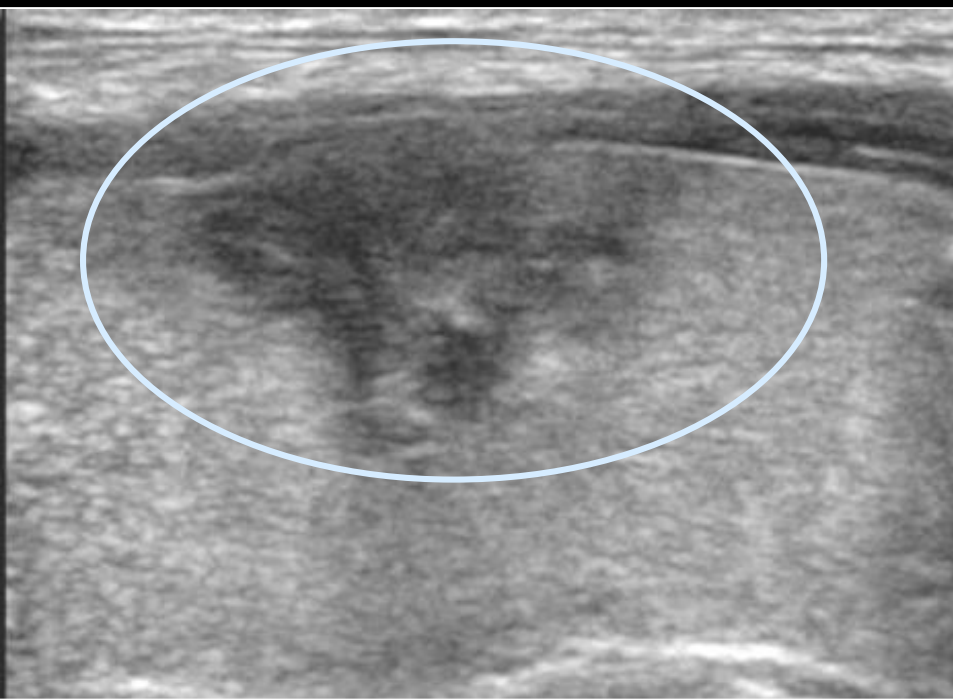
Papillary carcinoma



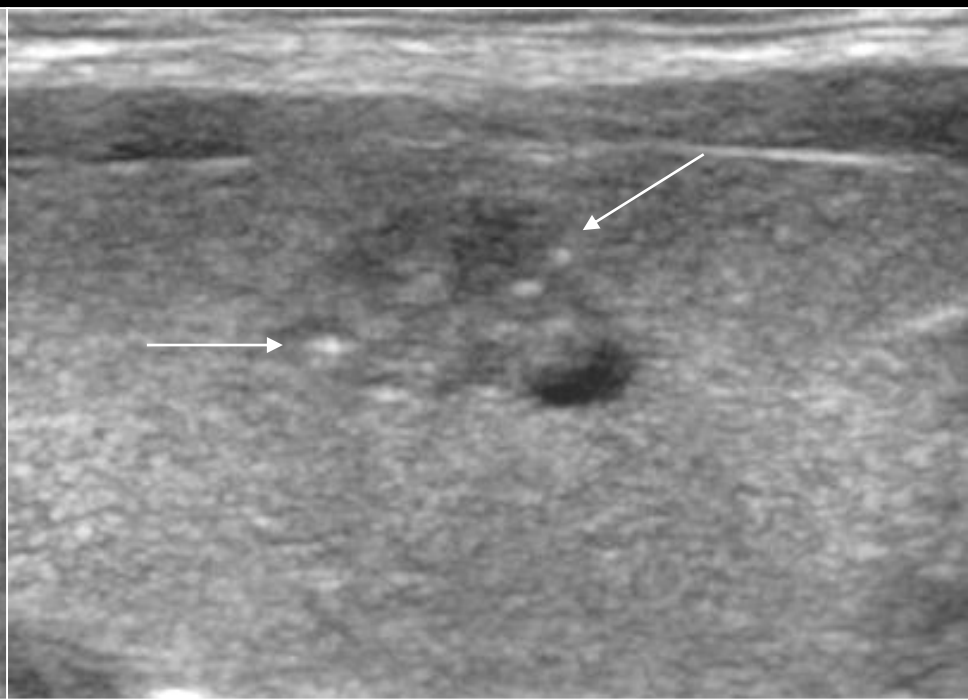
Longitudinal scan

- well-defined margin
- anechoic (cystic) portion 포함

Adenoma



Longitudinal scan



Longitudinal scan

- 1) Irregular margin
- 2) hypoechoic, solid nodule
- 3) Microcalcifications (arrows)

Papillary carcinoma

FEATURES		BENIGN	MALIGN
Internal content	Purely cystic content	++++	+
	Cystic with thin septa	++++	+
	Mixed solid and cystic mass	+++	++
	Comet tail artifacts	++++	+
Echogenicity	Hyperechoic	++++	+
	Isoechoic	+++	++
	Hypoechoic	+++	++++
Halo	Thin halo	++++	++
	Thick incomplete halo	+	+++
Margin	Well defined	+++	++
	Poorly defined	++	+++
Calcifications	Eggshell calcifications	++++	+
	Coarse calcifications	+++	++
	Micro-calcifications	++	++++
Doppler	Peripheral flow pattern	++++	++
	Internal flow pattern	++	++++

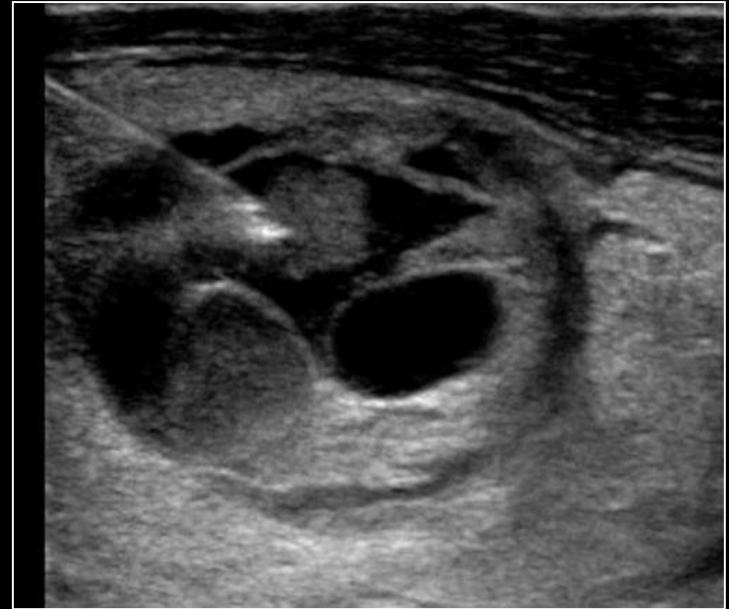
(요약1) US Criteria of Thyroid nodule

- TUS 1 (양성)

- cystic nodule
- Peripheral continuous halo
- heterogenous echo texture

- TUS 2 (양성추정)

- circumscribed margin
- iso-or hyperecho compared with strap muscle
- egg-shell calcification
- macrocalcification



(요약2) US Criteria of Thyroid Nodule

TUS 3 (미결정)

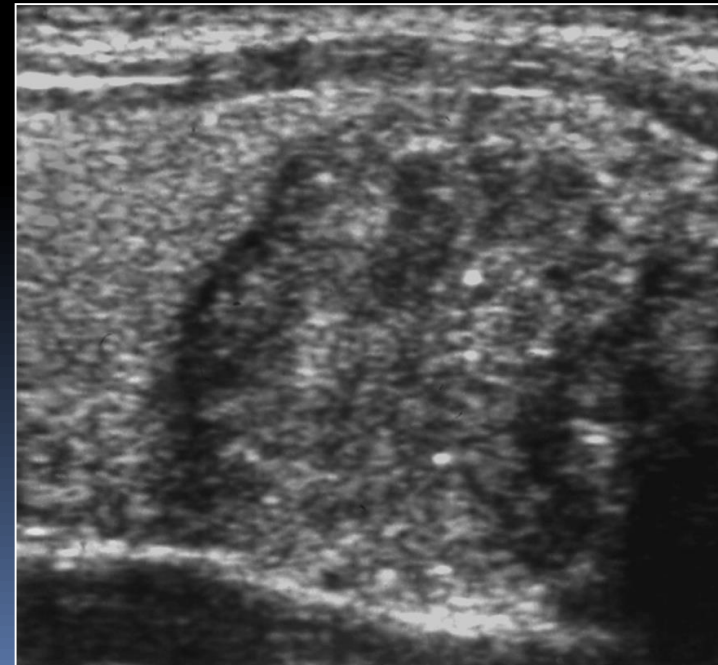
- hypoecho compared with adjacent thyroid
- no cystic change
- well-defined margin

TUS 4 (악성 추정)

- homog. hypoecho compared with adjacent strap muscle
- ill-defined margin

TUS 5 (강한 악성)

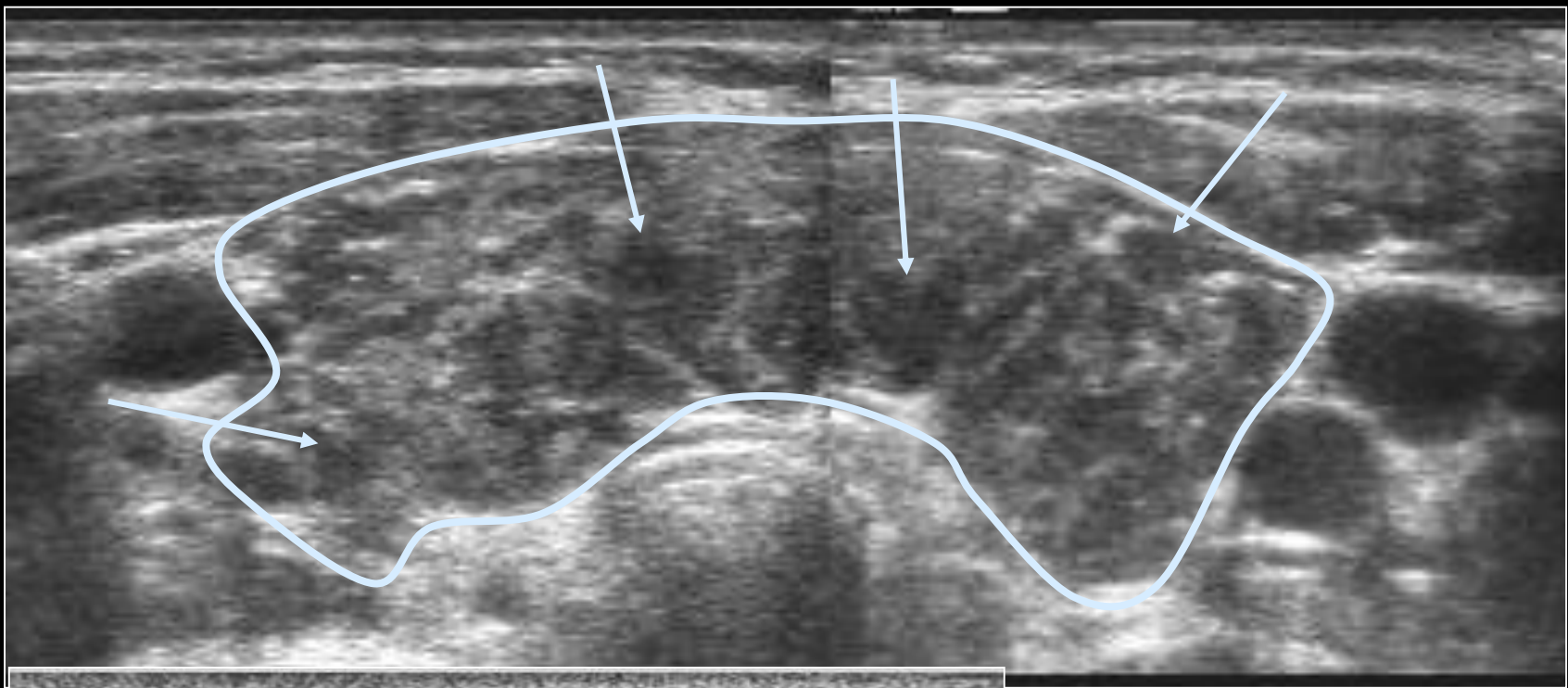
- punctate microcalcification,
- ill-defined, margin
- A-P diameter > Rt-Lt diameter,
- Lymph node (+)





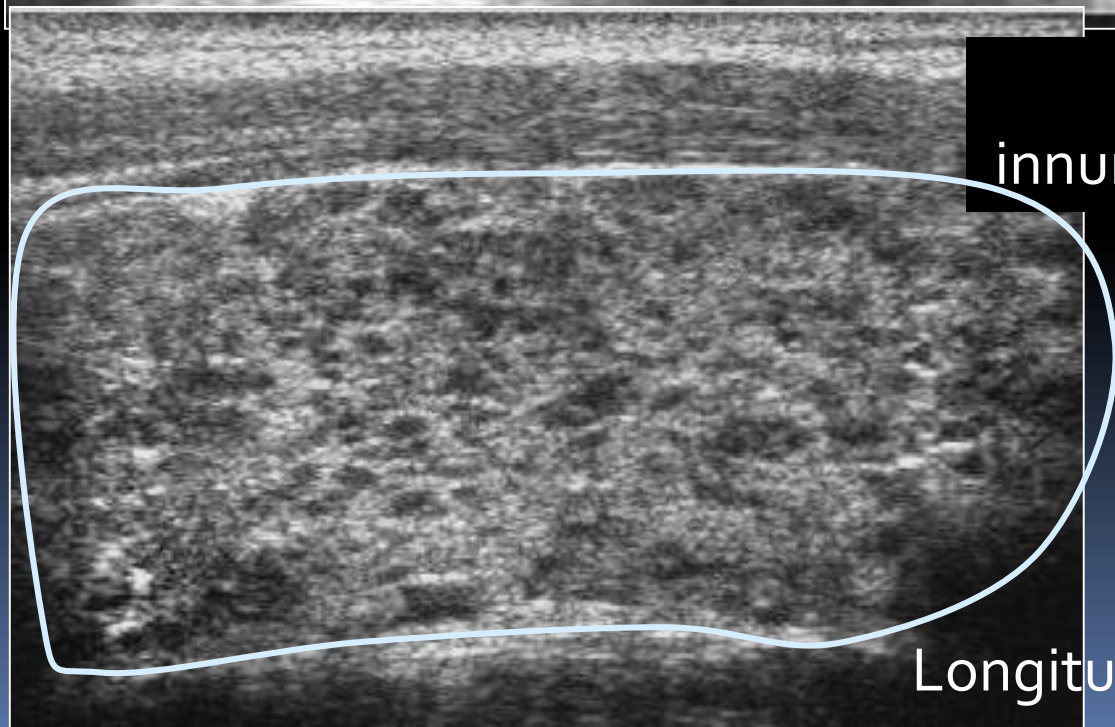
Diffuse thyroid disease

- Hashimoto thyroiditis
 - Lymphocytic thyroiditis
- 



Diffuse involvement
innumerable hypoechoic nodules

Diffuse thyroid disease

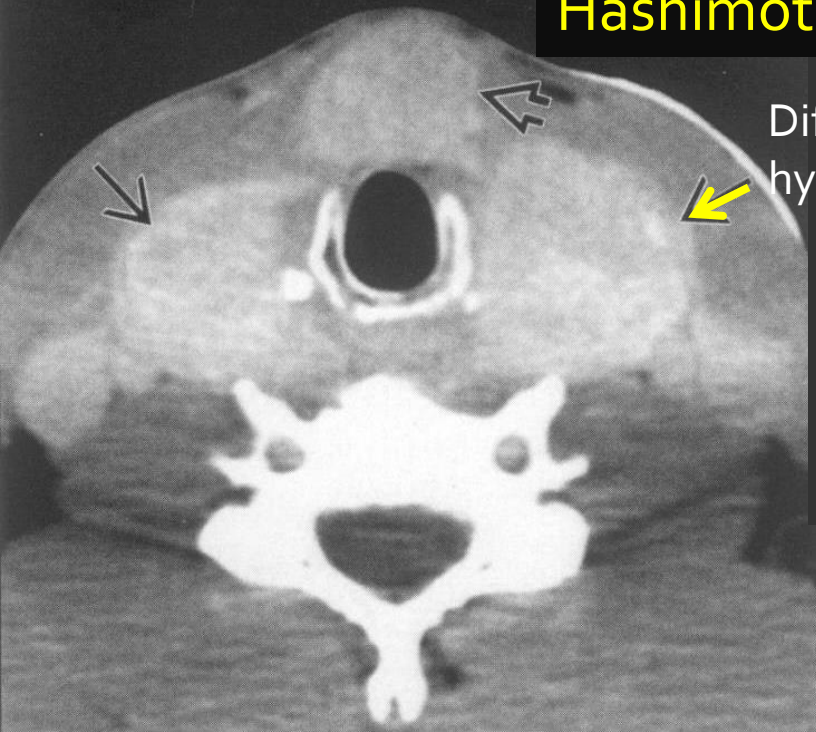


Longitudinal scan

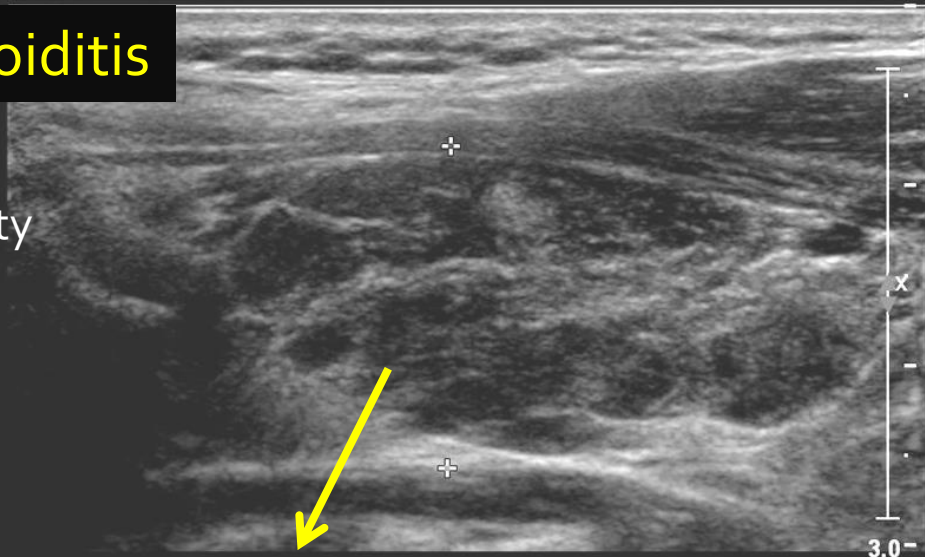
Hashimoto Thyroiditis

- Chronic, autoimmune-mediated lymphocytic inflammation (=IgG₄-related disease)
- Early stage: enlargement
- Late stage: atrophy
- CT
 - Diffuse decreased density typical
 - No necrosis or calcification
- US
 - lobulated, decreased echo
 - Micronodulation

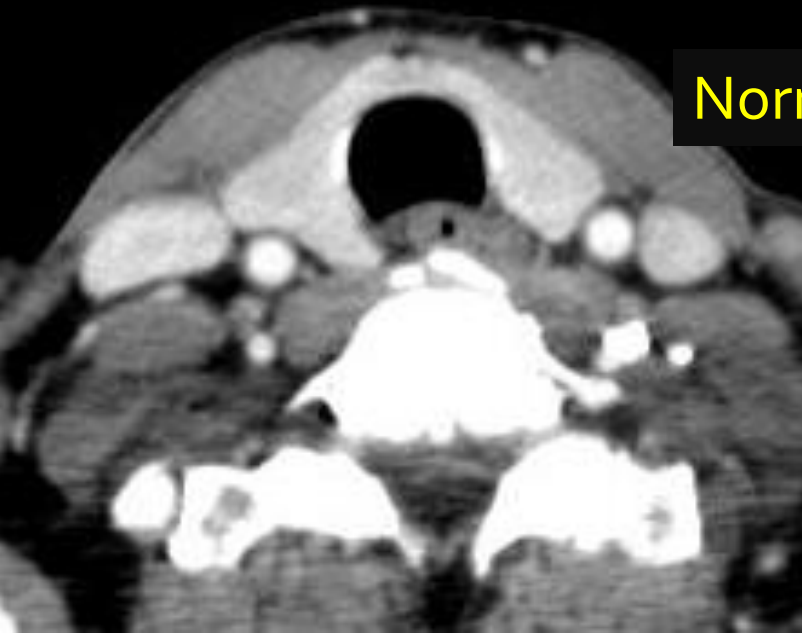
Hashimoto thyroiditis



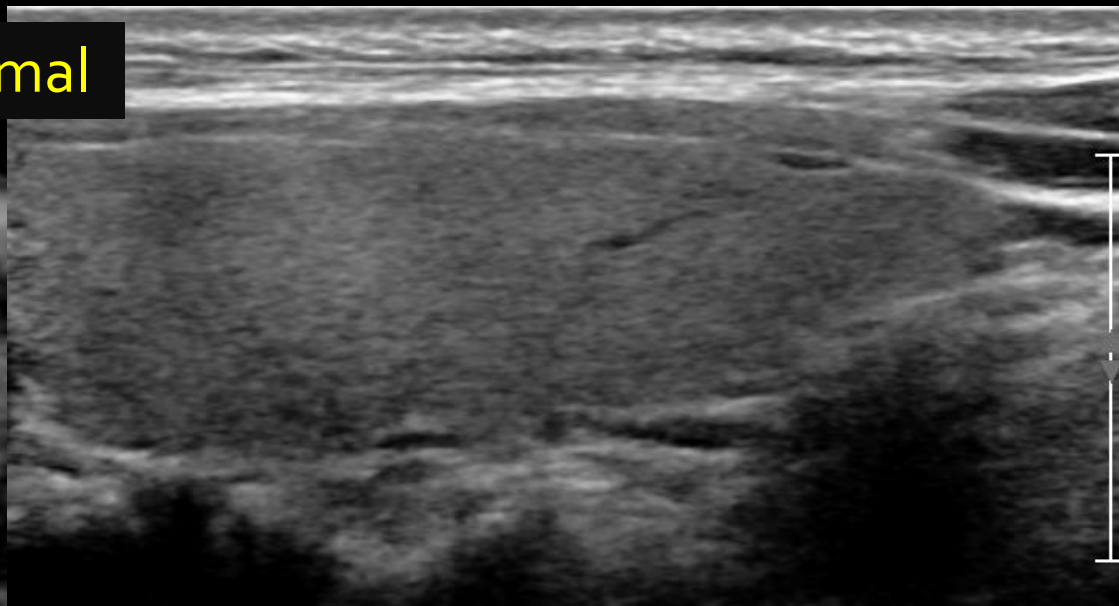
Diffuse
hypodensity

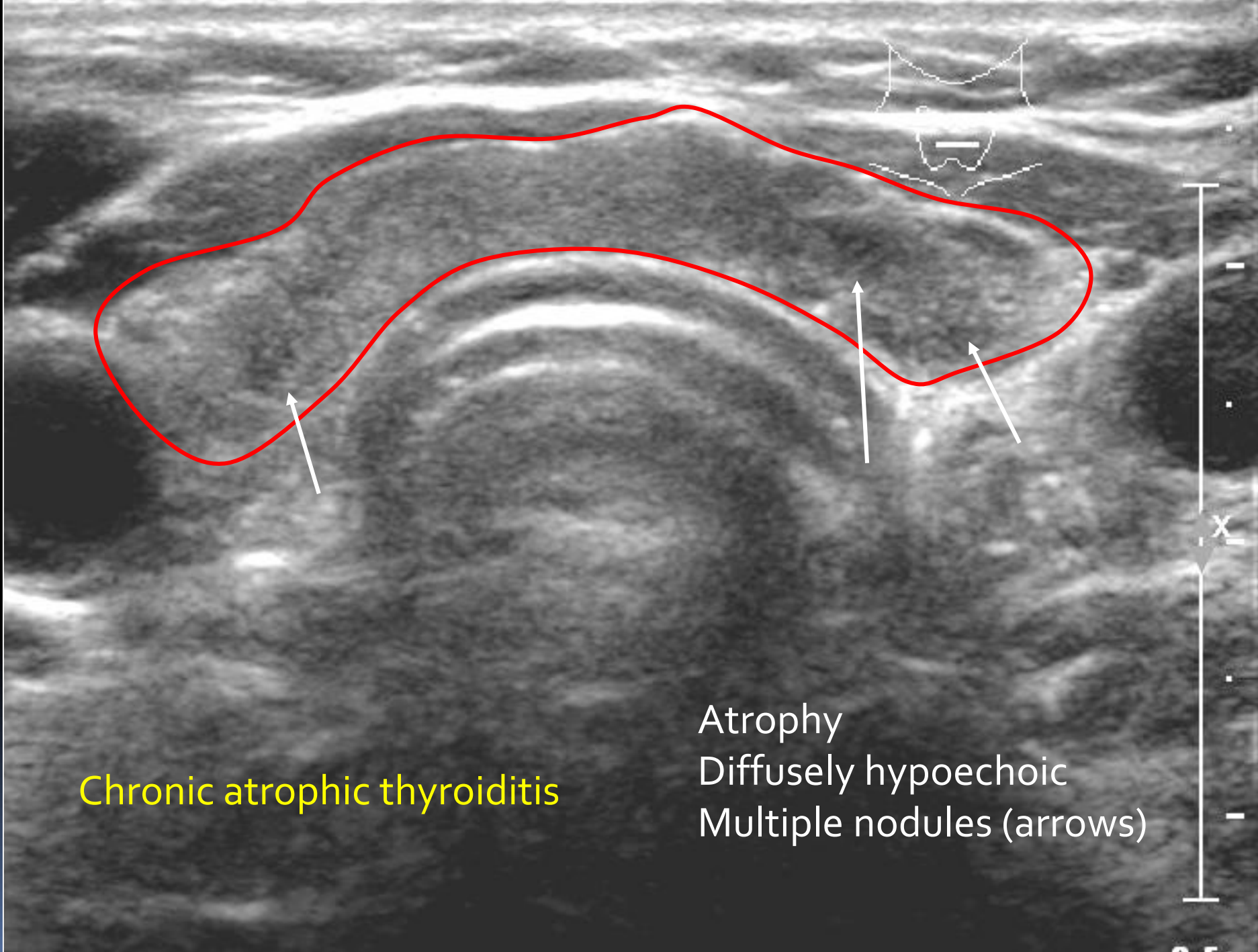


Diffusely hypoechoic
Lobulated



Normal

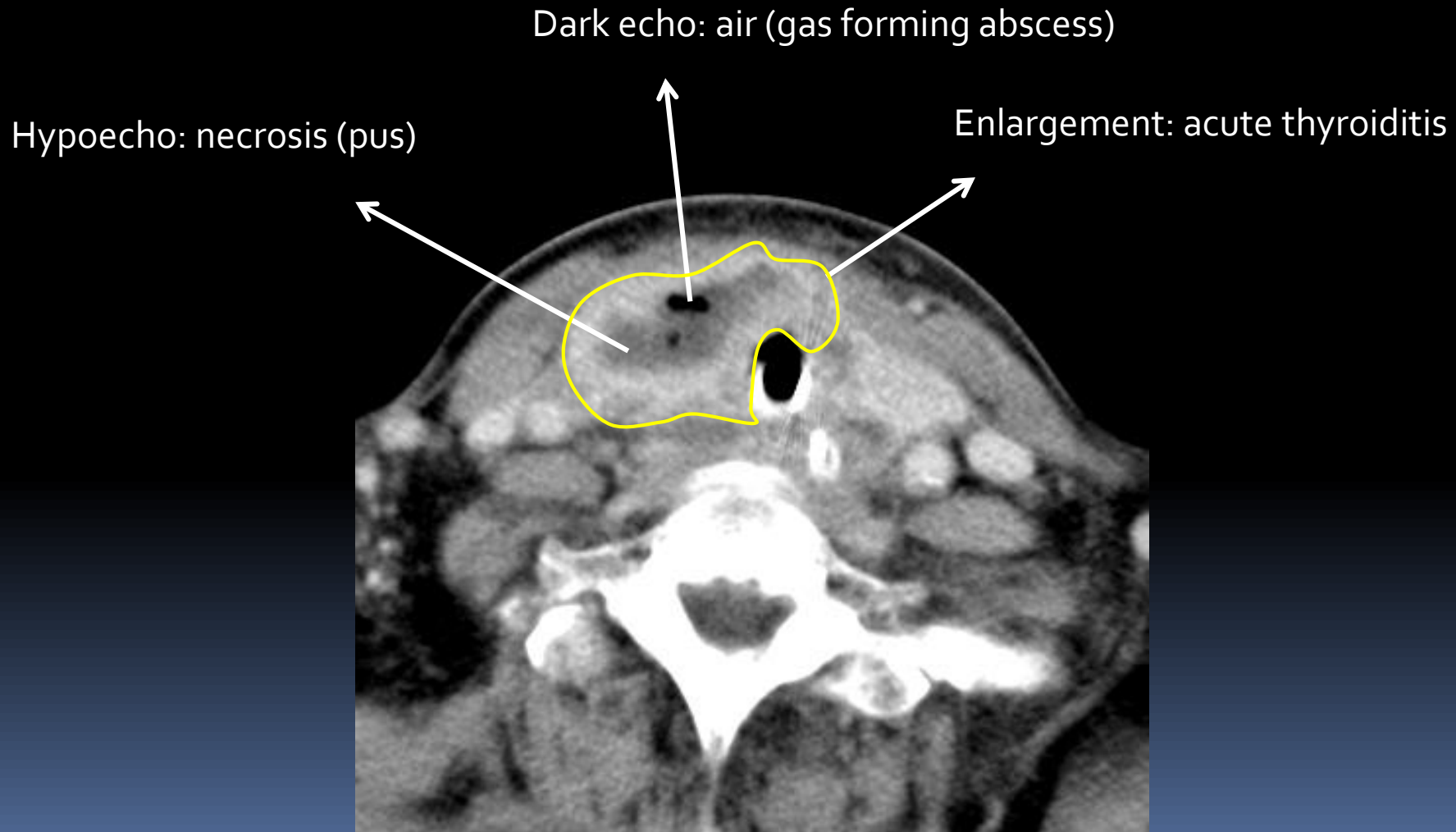




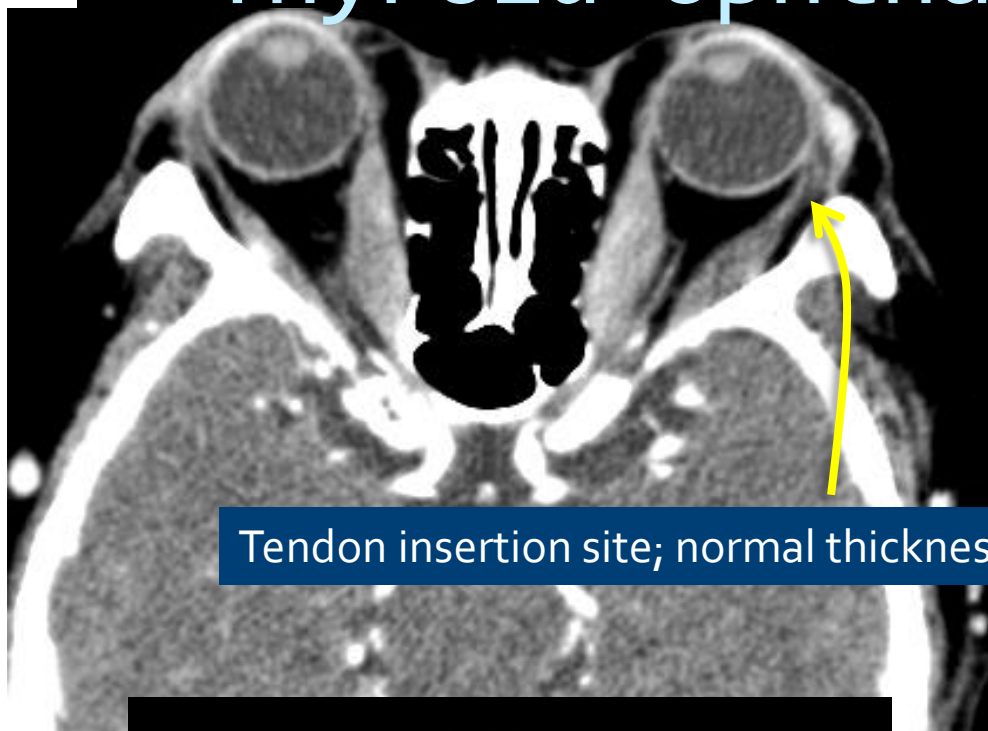
Chronic atrophic thyroiditis

Atrophy
Diffusely hypoechoic
Multiple nodules (arrows)

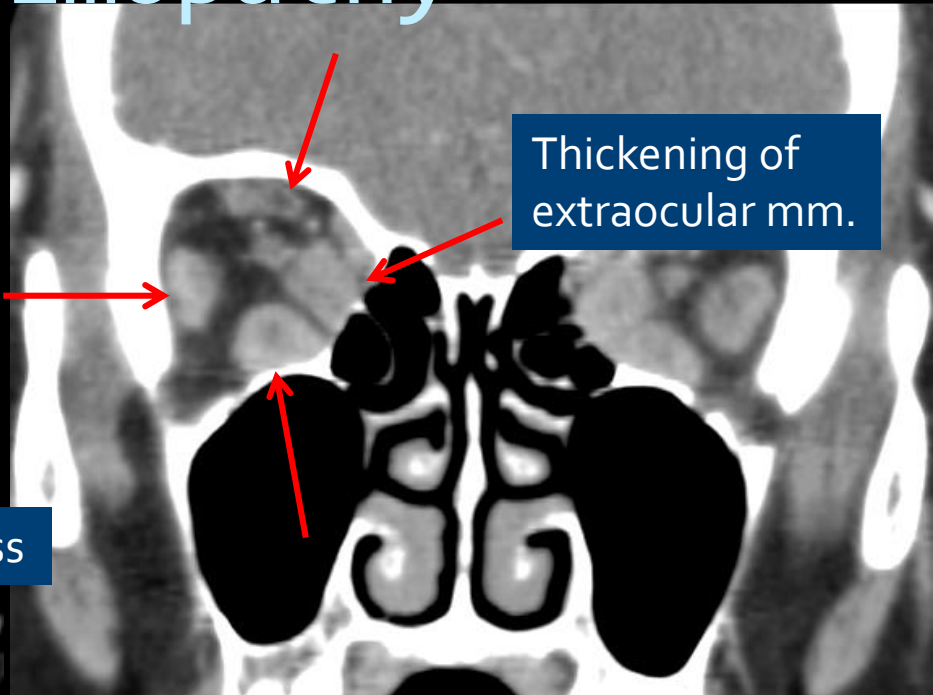
Thyroid abscess



Thyroid ophthalmopathy



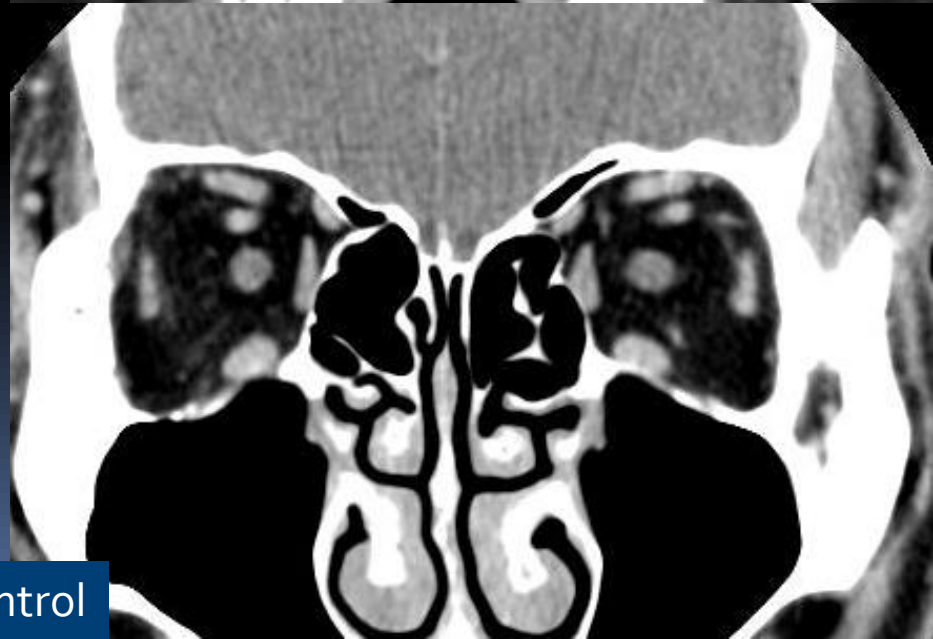
Tendon insertion site; normal thickness



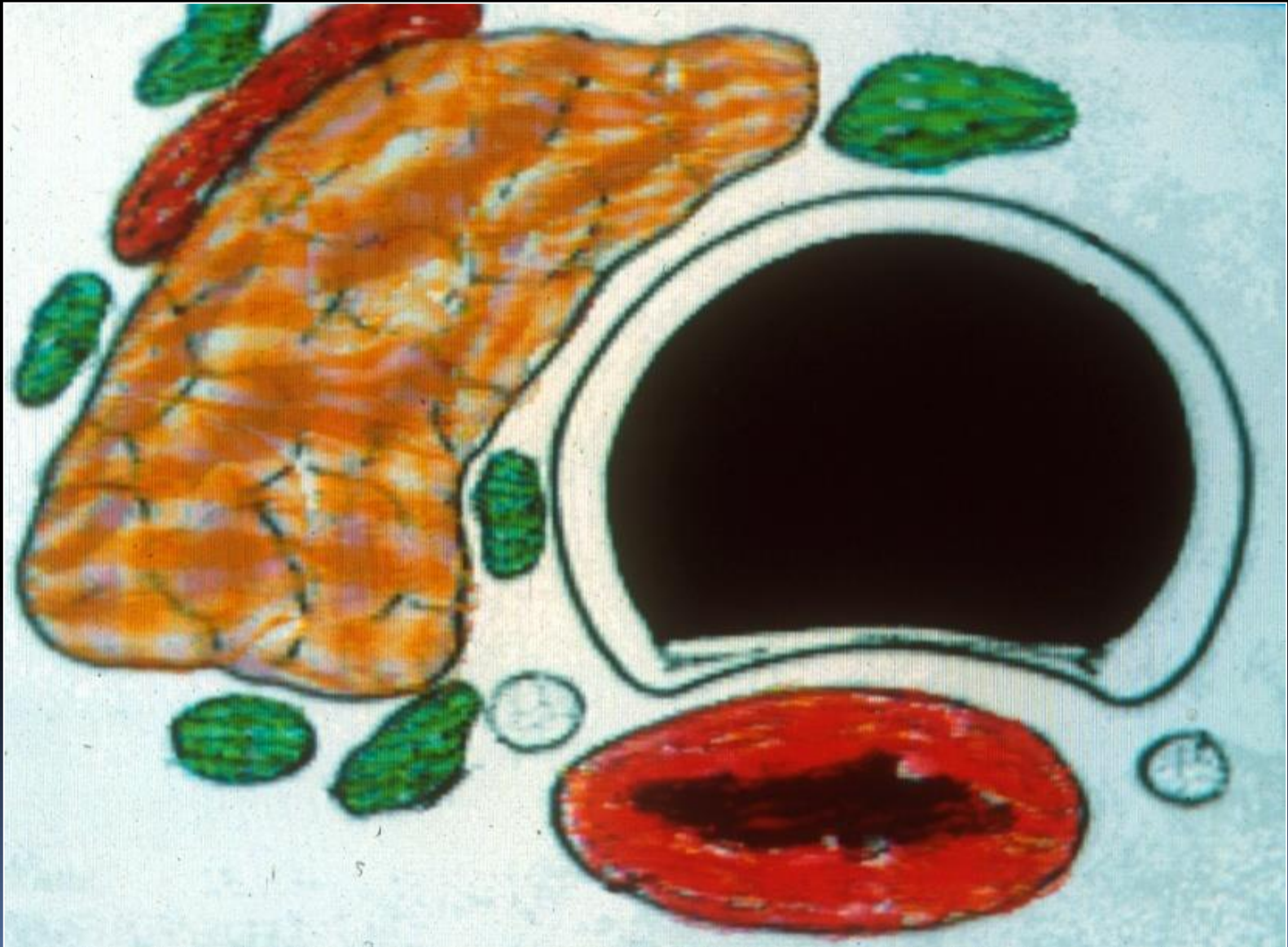
Thickening of extraocular mm.



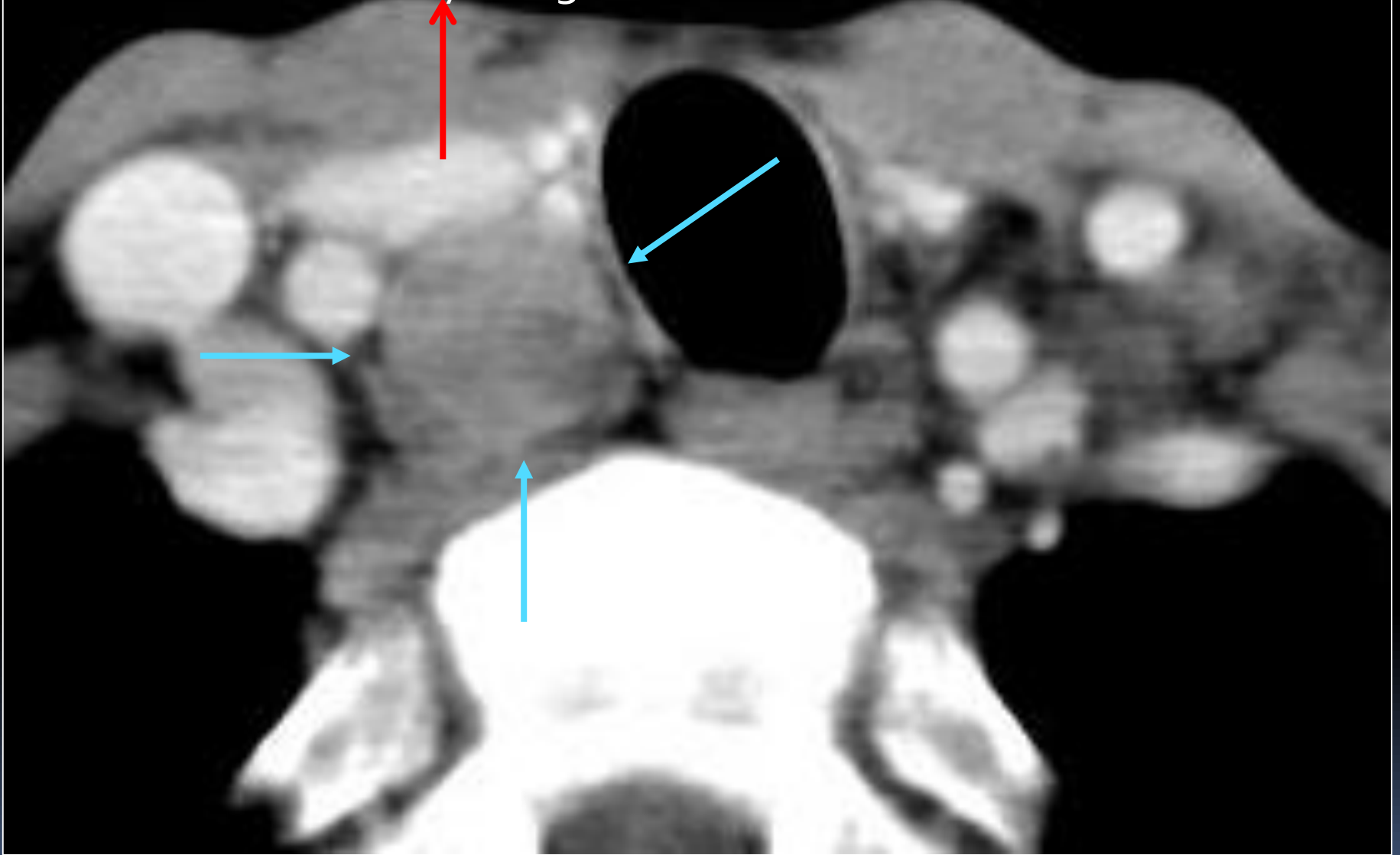
Normal control



PARATHYROID GLANDS



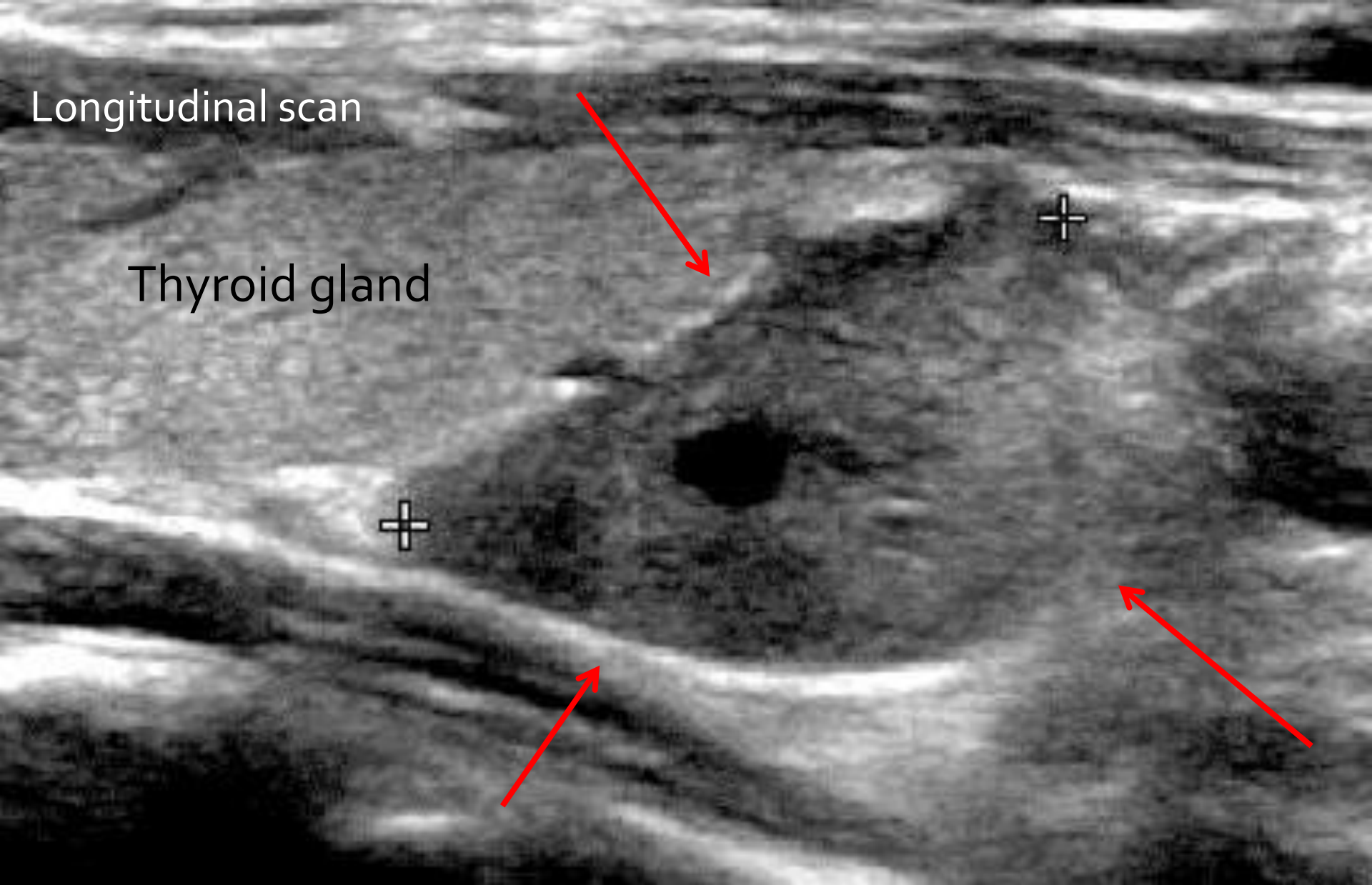
Thyroid gl.



Parathyroid carcinoma

Longitudinal scan

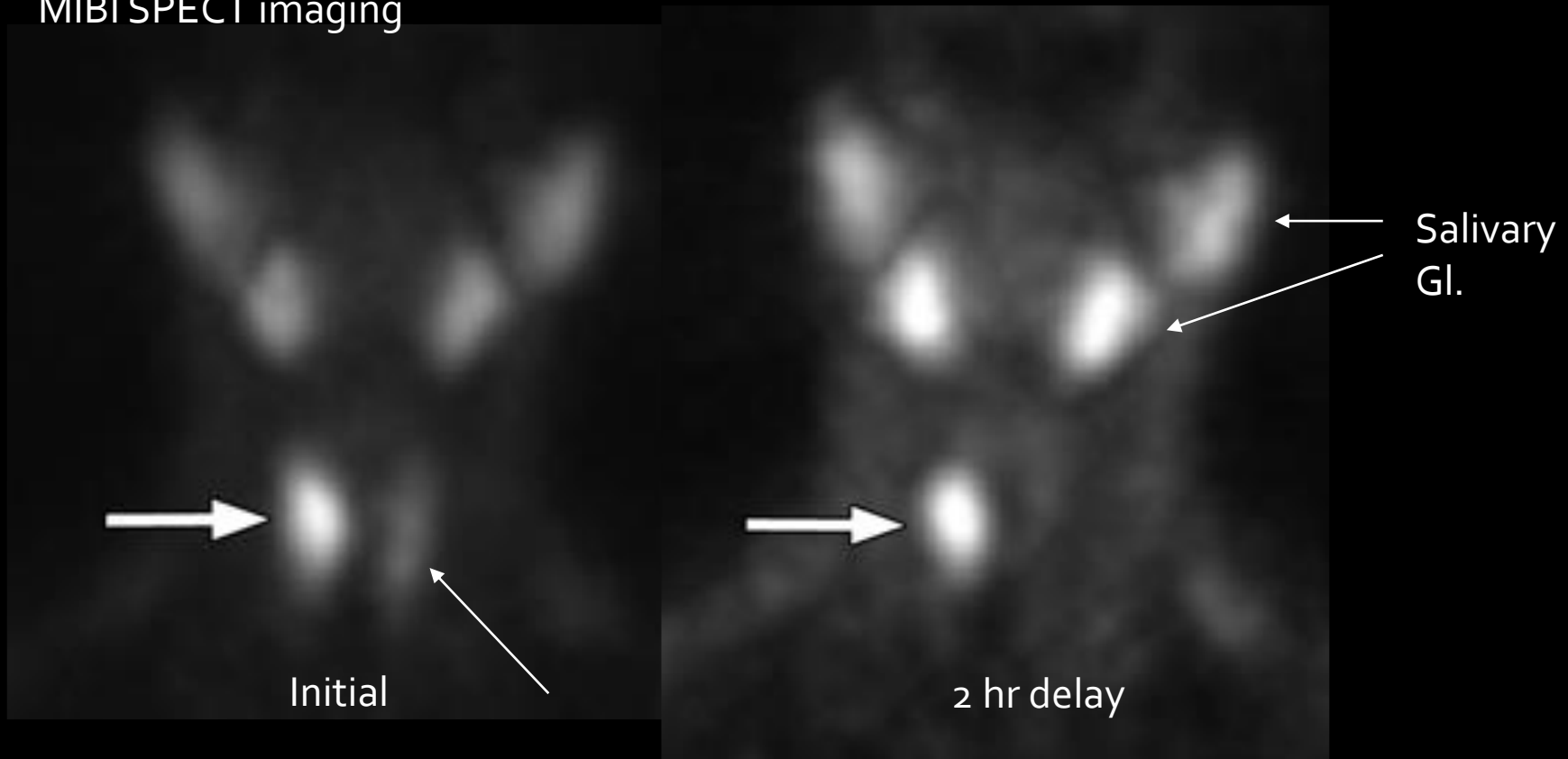
Thyroid gland



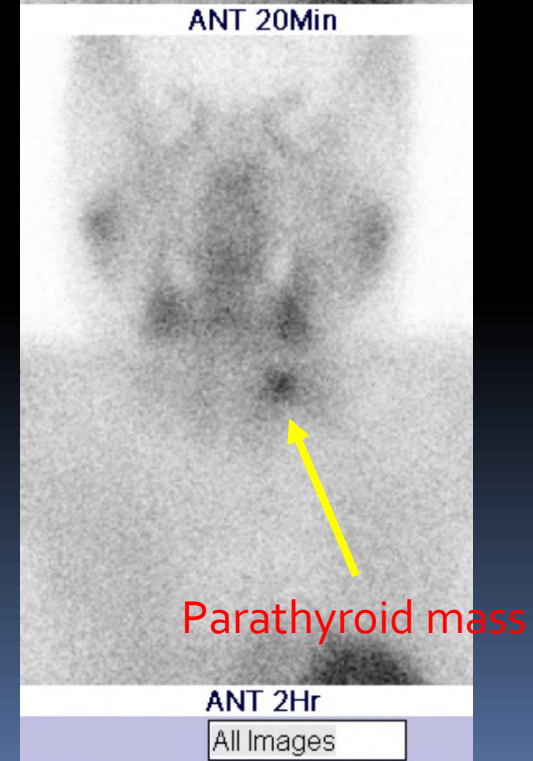
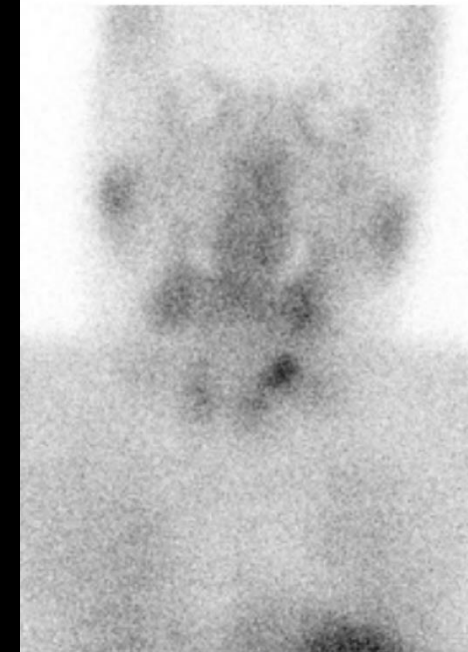
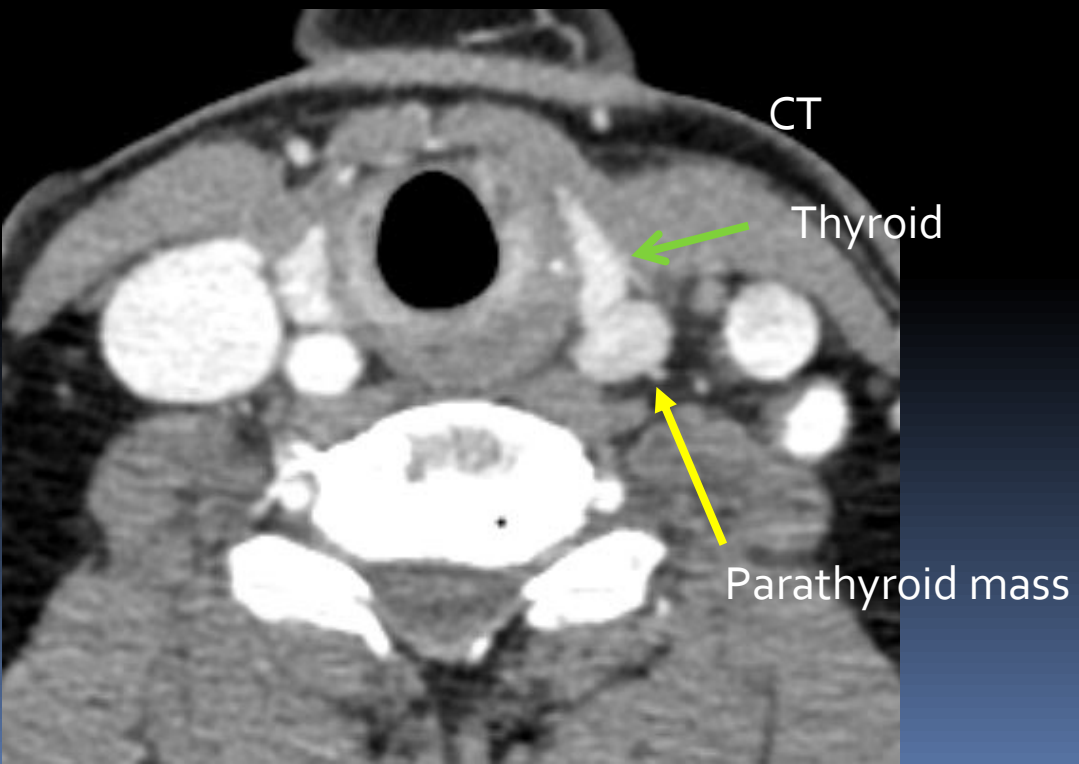
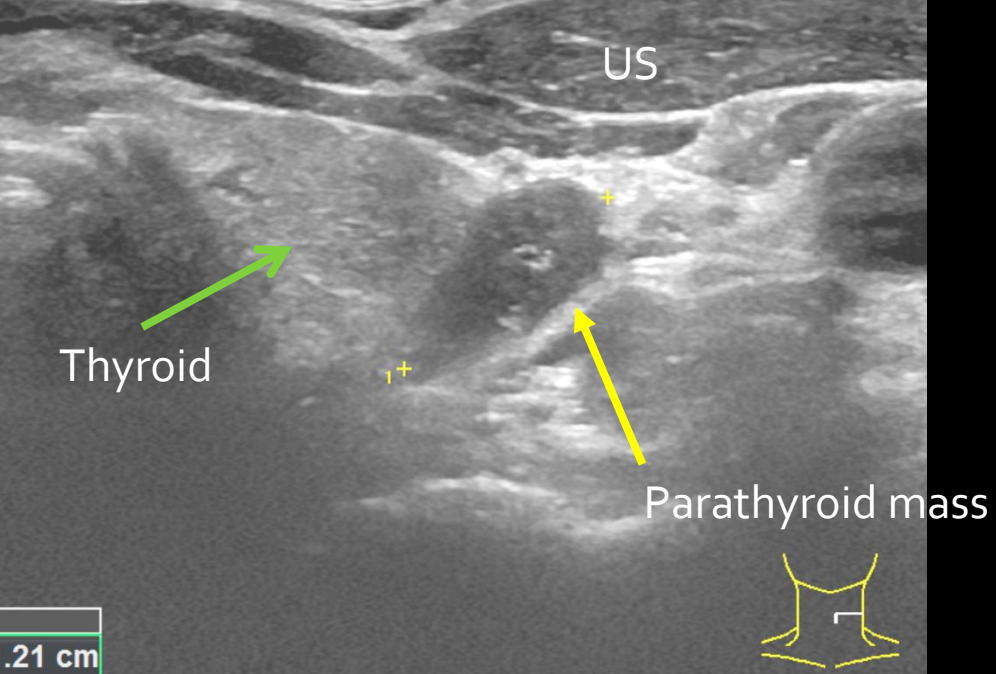
Parathyroid adenoma

Well-defined
Hypoechoic solid nodule

MIBI SPECT imaging



- Left thyroid gland: 2 hr delay image에서 보이지 않음(정상)
Right parathyroid mass: 2 hr delay image에서 더 뚜렷이 보임
- parathyroid adenoma는 TC99m MIBI를 강하게 결합
 - Salivary gland도 later stage에 정상적으로 radiotracer 보임.





Imaging Recommendations

- *Ultrasonography*
 - DDx: benign vs malignant nodules
 - for multiple nodules
 - for non-palpable nodules
 - for fine needle aspiration guidance
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학생들이 알아야 할 내용

- CT, MRI, US: 정상 해부학
- 병변의 유무
- 병변의 형태 분별
 - 경계
 - 내용물 (solid vs cystic)
 - 조영증강
- 감별진단
 - Benign vs malignant